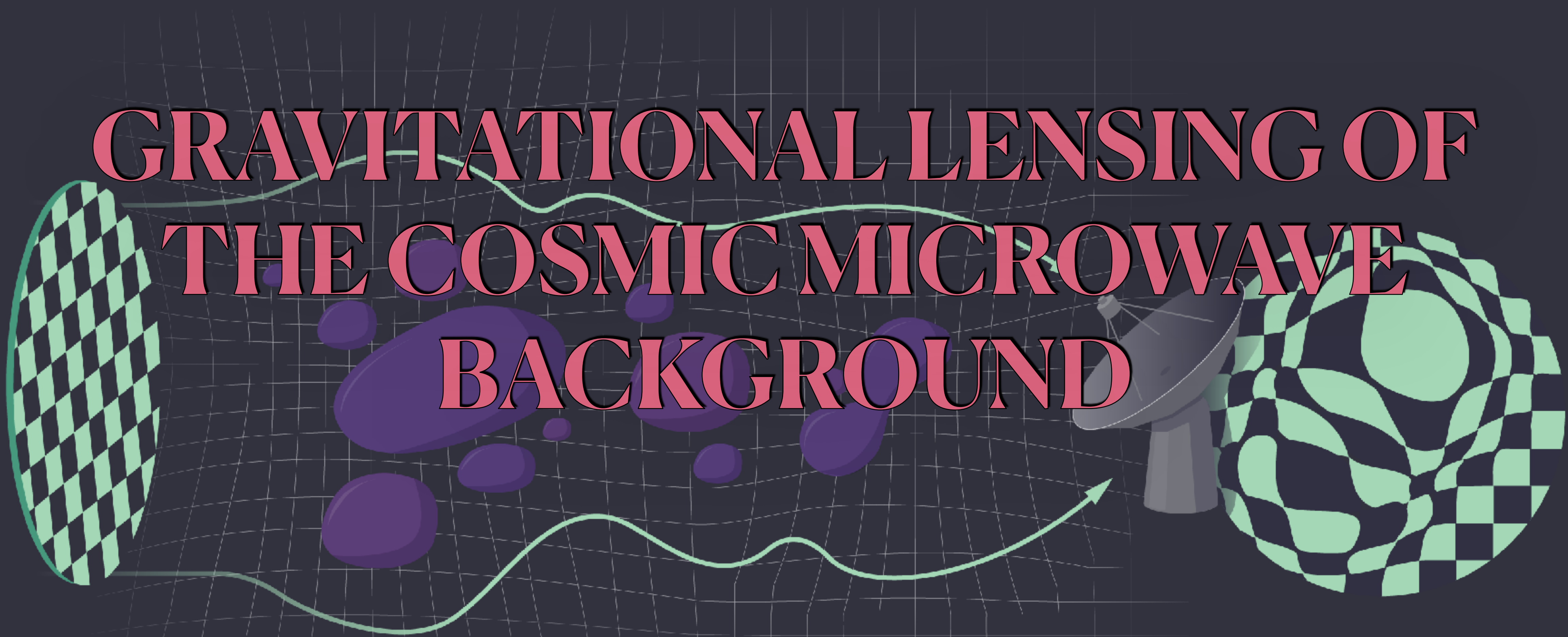


GRAVITATIONAL LENSING OF THE COSMIC MICROWAVE BACKGROUND



Irene Abril Cabezas, 2026 Flatiron CMB Summer School

Outline

- 1. Motivation**
- 2. CMB lensing basics**
- 3. Effect on the CMB**
- 4. Delensing**
- 5. Lensing reconstruction**
- 6. Lensing power spectrum measurement and challenges**

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DISCLAIMER: None of the ideas my own. This is what I have learned from many, including Blake Sherwin, (Hanson &) Lewis & Challinor, **Frank Qu**, Sigurd Naess, Jo Dunkley, Hu & Okamoto, Mat Madhavacheril, Alex van Engelen, Toshiya Namikawa

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llenges

Why study CMB lensing?

“Observable that has emerged as a robust and powerful probe for Cosmology”

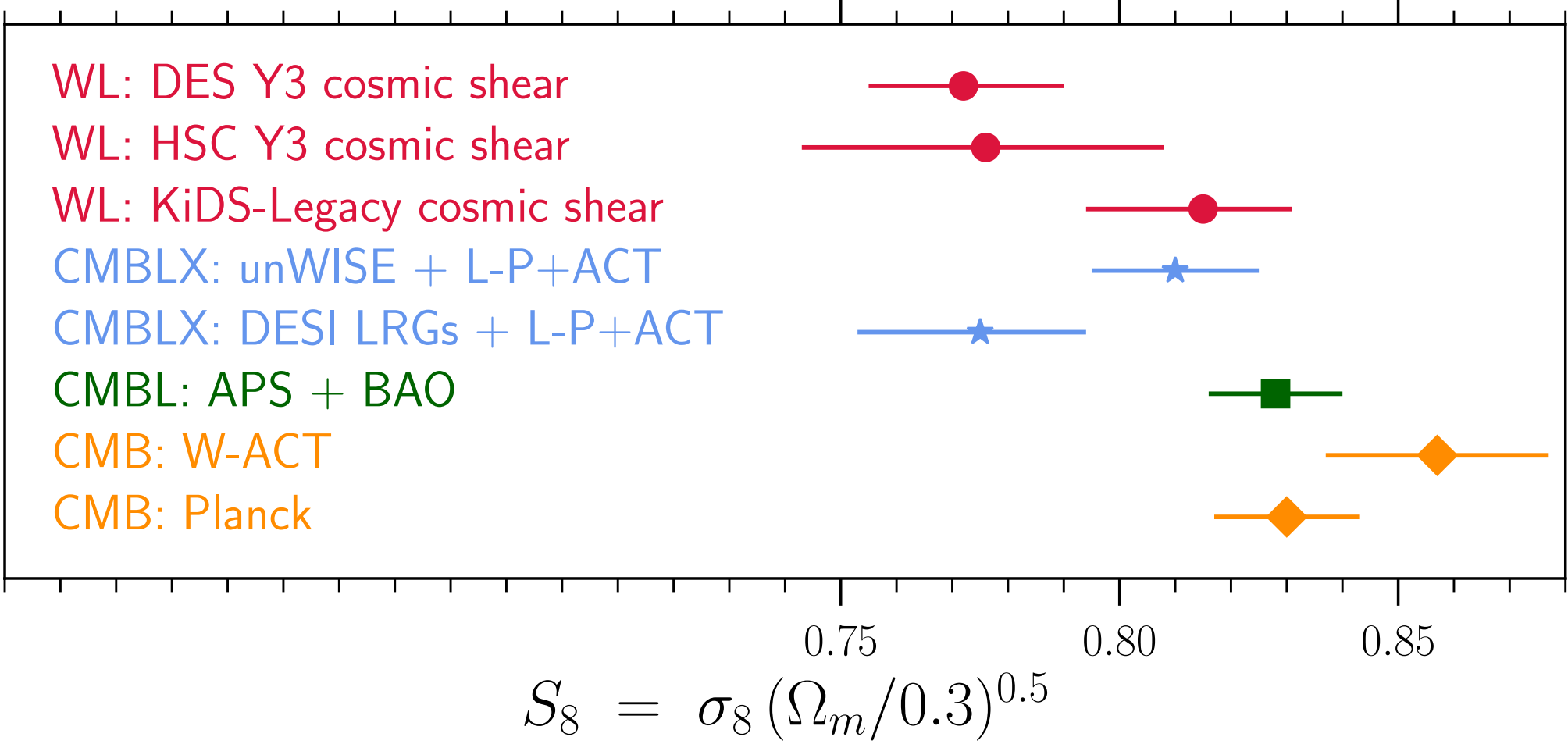
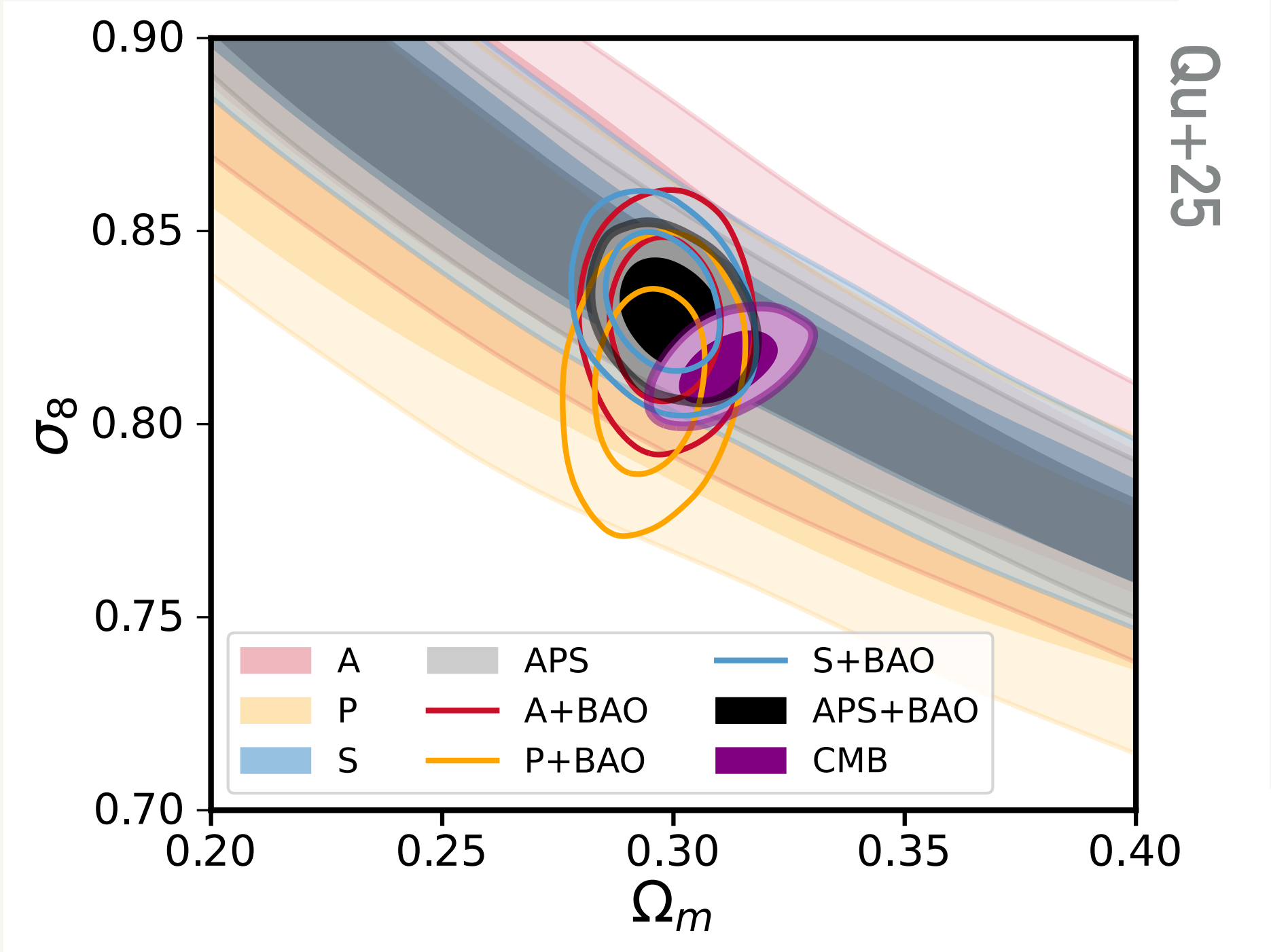
As a nuisance:

- It has a small but non-negligible effect on the **CMB power spectra**
- Lensing effects generate a curl-like (B-mode) polarisation pattern that acts as confusion noise for CMB polarisation experiments targeting a **primordial gravitational wave signal**

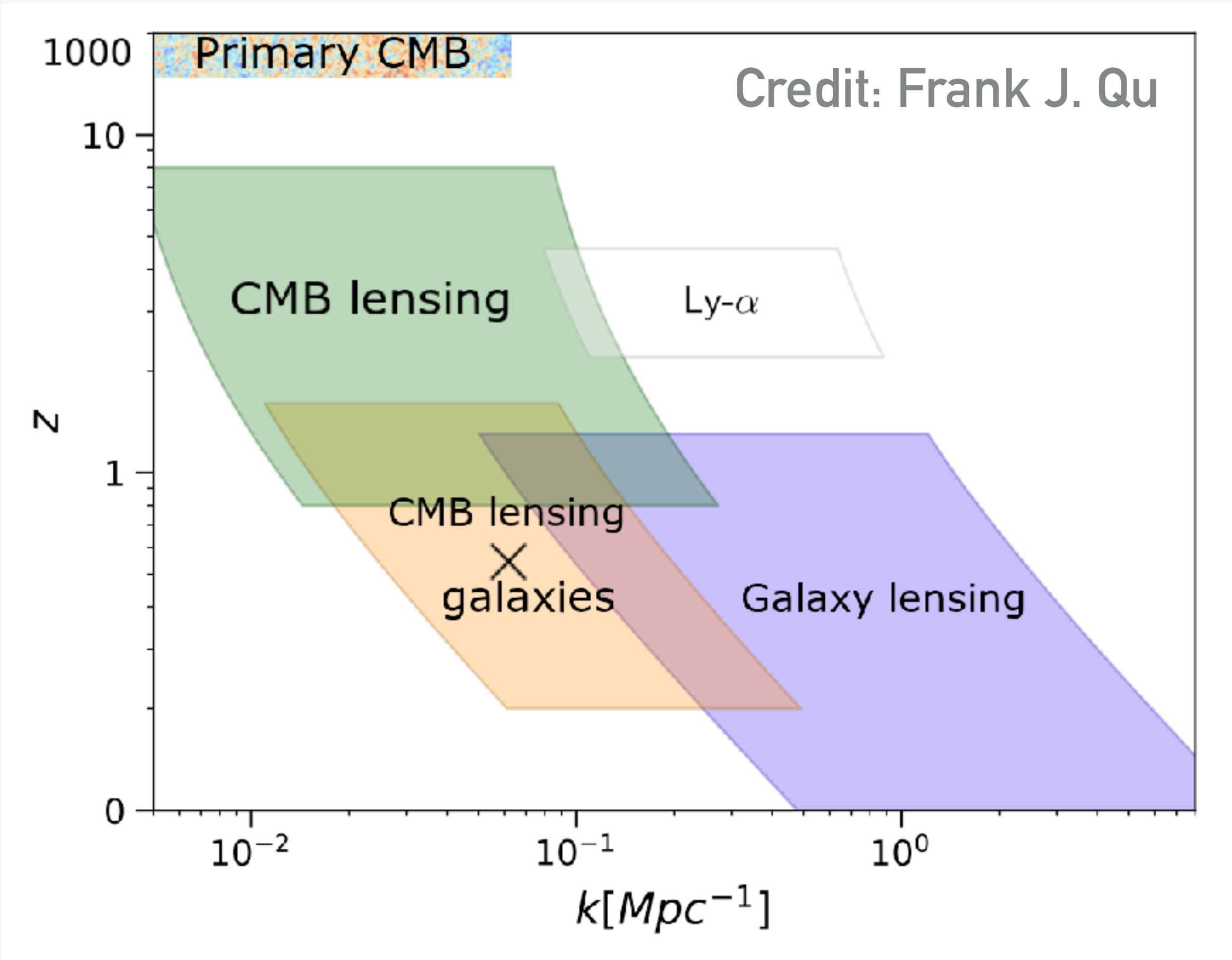
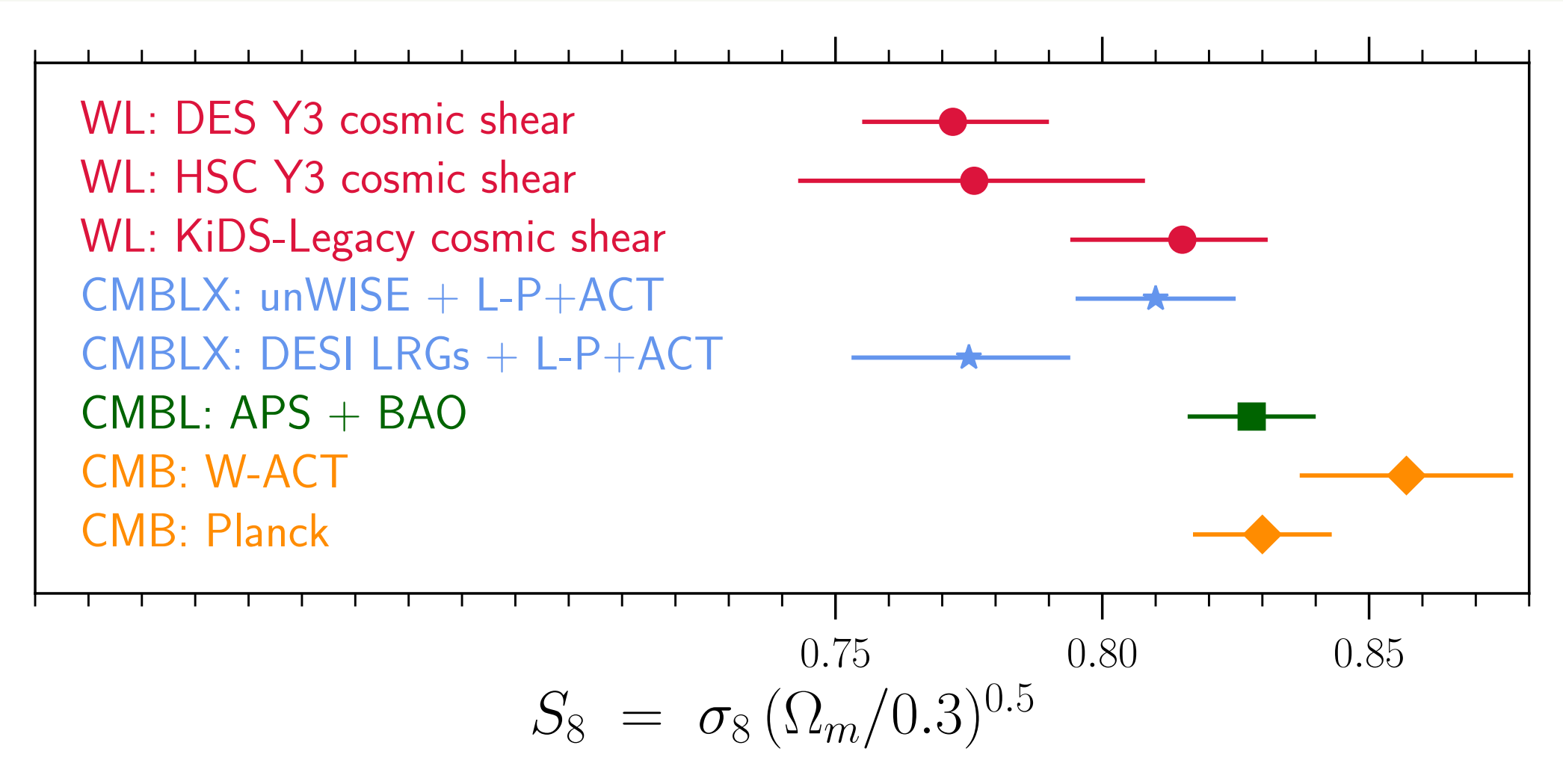
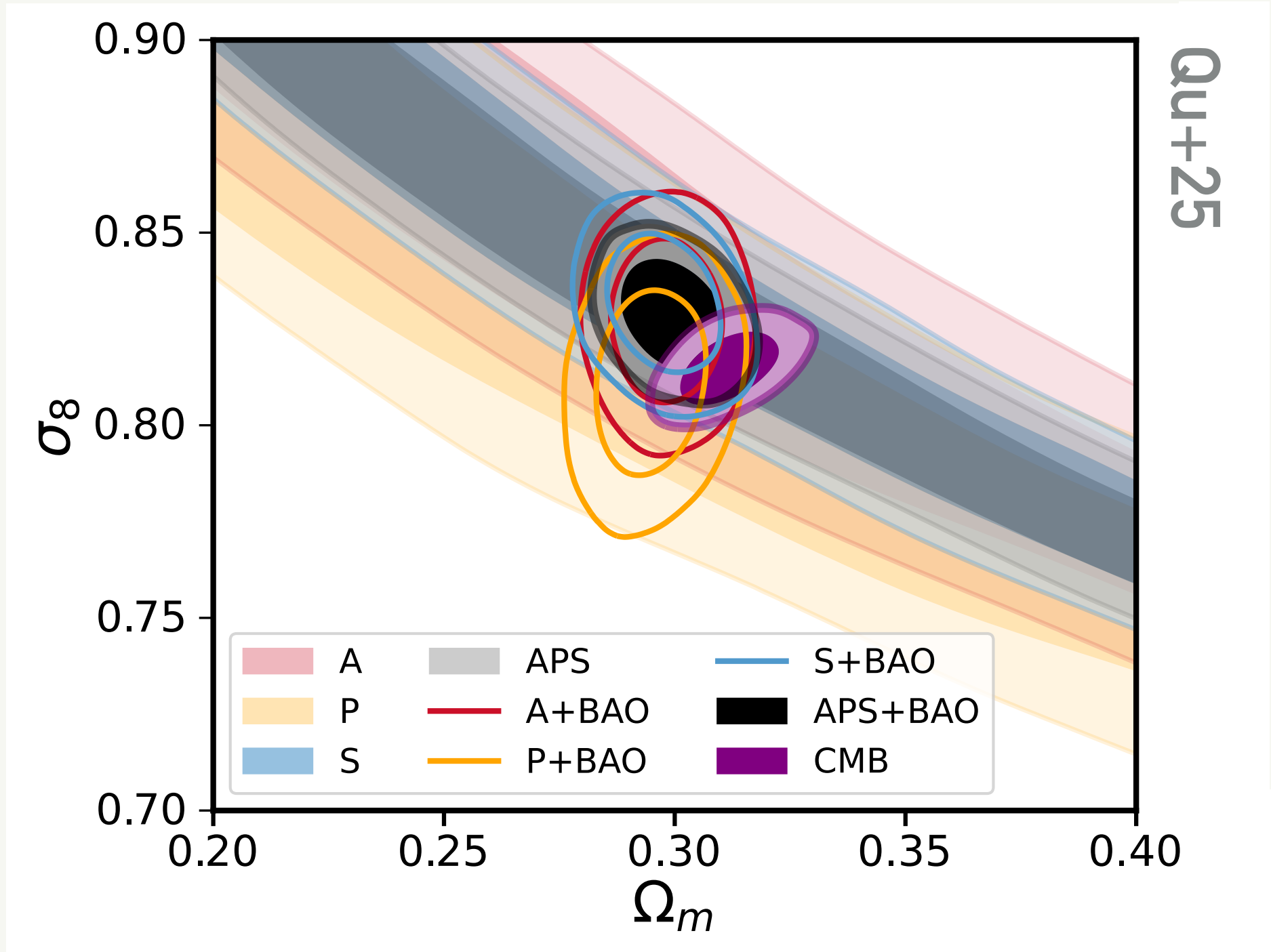
Just for being itself:

- It probes the large-scale matter distribution in the Universe:
 - How do cosmic structures growth? (σ_8)
 - What are the properties/masses of neutrinos? ($\sum m_\nu$)

Structure growth over time as a powerful probe of new physics



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CMB lensing

order of magnitude + geometry

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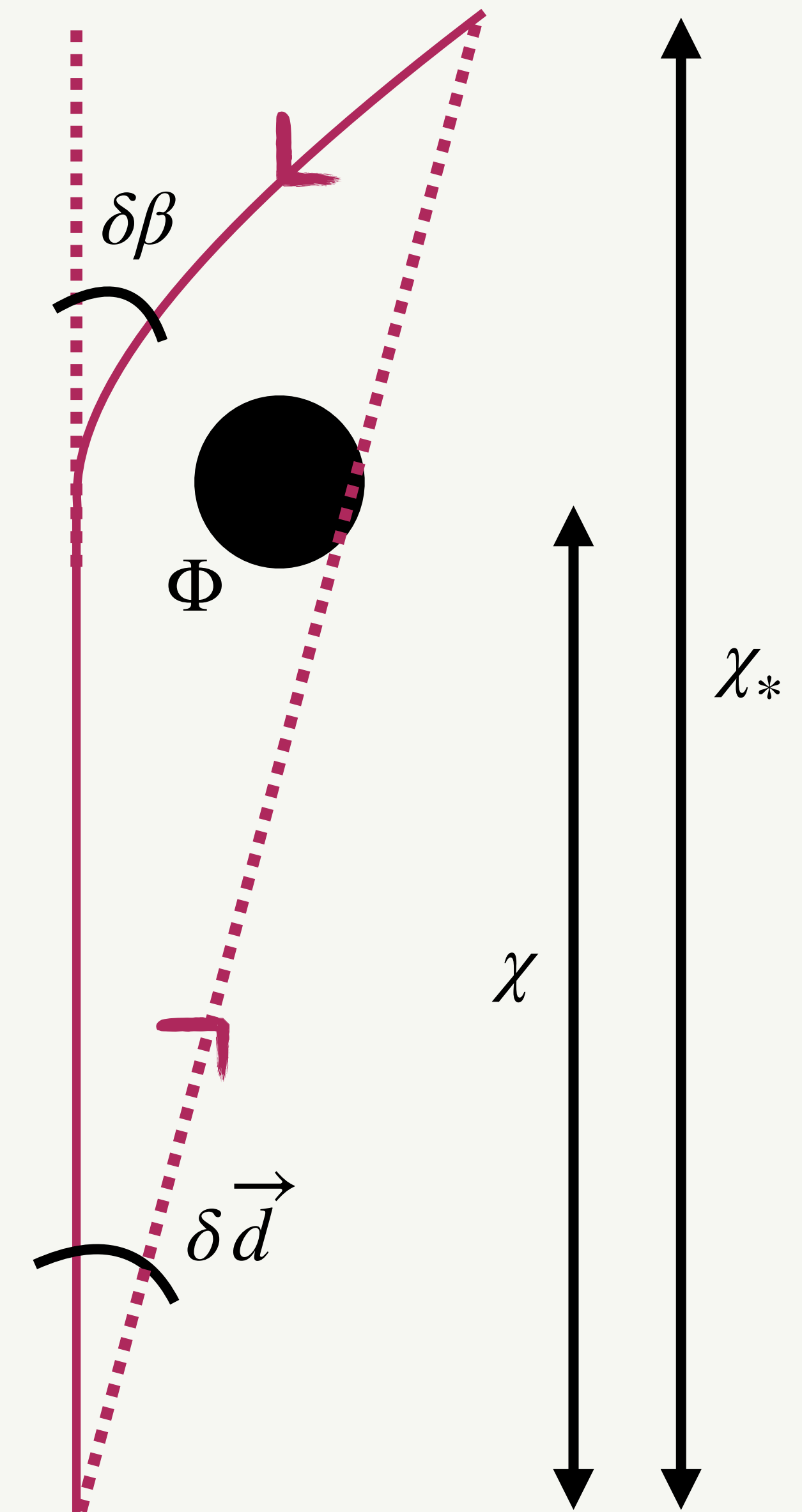
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$$\delta\beta = -2\delta\chi \nabla_{\perp} \Phi \quad \Rightarrow \quad \delta \vec{d} = -2 \left(\frac{\chi_* - \chi}{\chi_*} \right) \delta\chi \nabla_{\perp} \Phi$$



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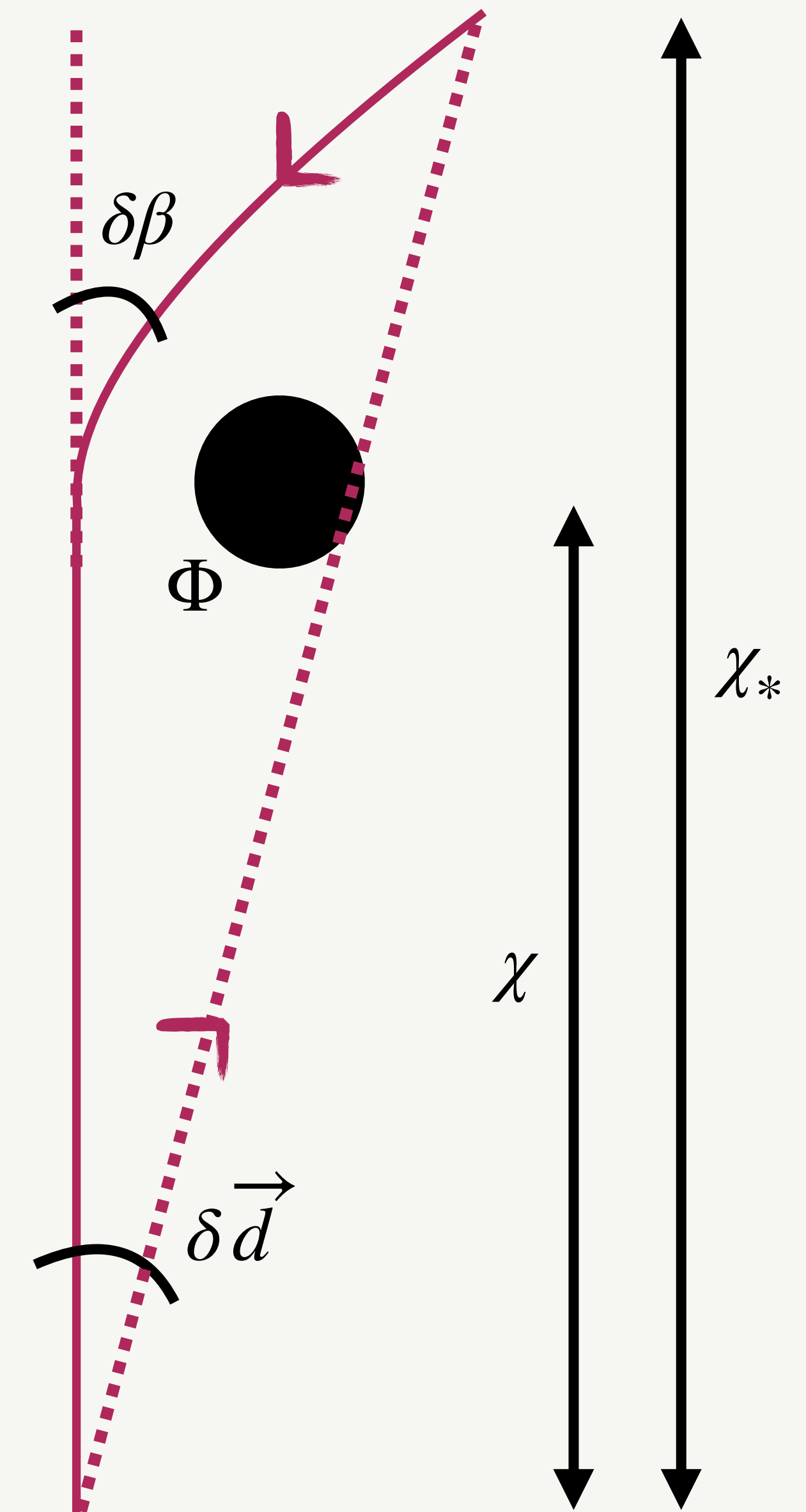
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- Summing up all small deflections along photon's path:

$$\vec{d}(\hat{n}) = -2 \int_0^{\chi_*} d\chi \left(\frac{\chi_* - \chi}{\chi_*} \right) \nabla_{\perp} \Phi$$



The lensing potential

And other ways to present equivalent information $(\vec{d}, \phi, \kappa)$

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The lensing potential is a projection of the 3D gravitational potentials in our Universe (with some distance weighting):

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Relationship between angular gradient and the actual 3D perpendicular gradient: $\nabla = \partial_\theta = \chi \nabla_\perp$

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How do we probe the projected total mass density in each direction?

$$\kappa \equiv -\frac{1}{2} \nabla^2 \phi \simeq \int_0^{\chi_*} d\chi \frac{\chi(\chi_* - \chi)}{\chi_*} \nabla_{3D}^2 \Phi = \Omega_{m,0} \int_0^{\chi_*} d\chi \tilde{W}(\chi) \delta(\hat{n}, \chi)$$

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Poisson's equation:

$$\nabla_{3D}^2 \Phi = \frac{3}{2a} \Omega_{m,0} H_0^2 \delta$$

Geometric projection kernel:

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κ is the quantity that directly probes the mass distribution in our Universe

fractional mass over-density

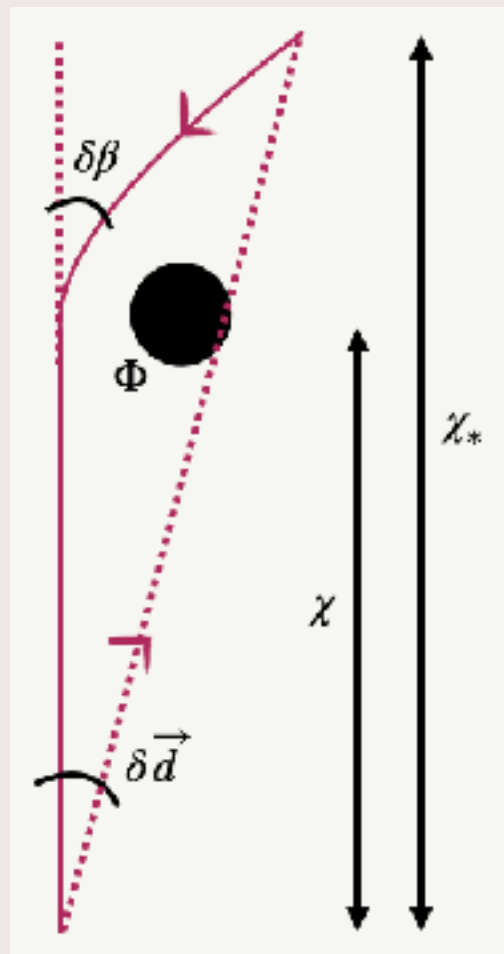
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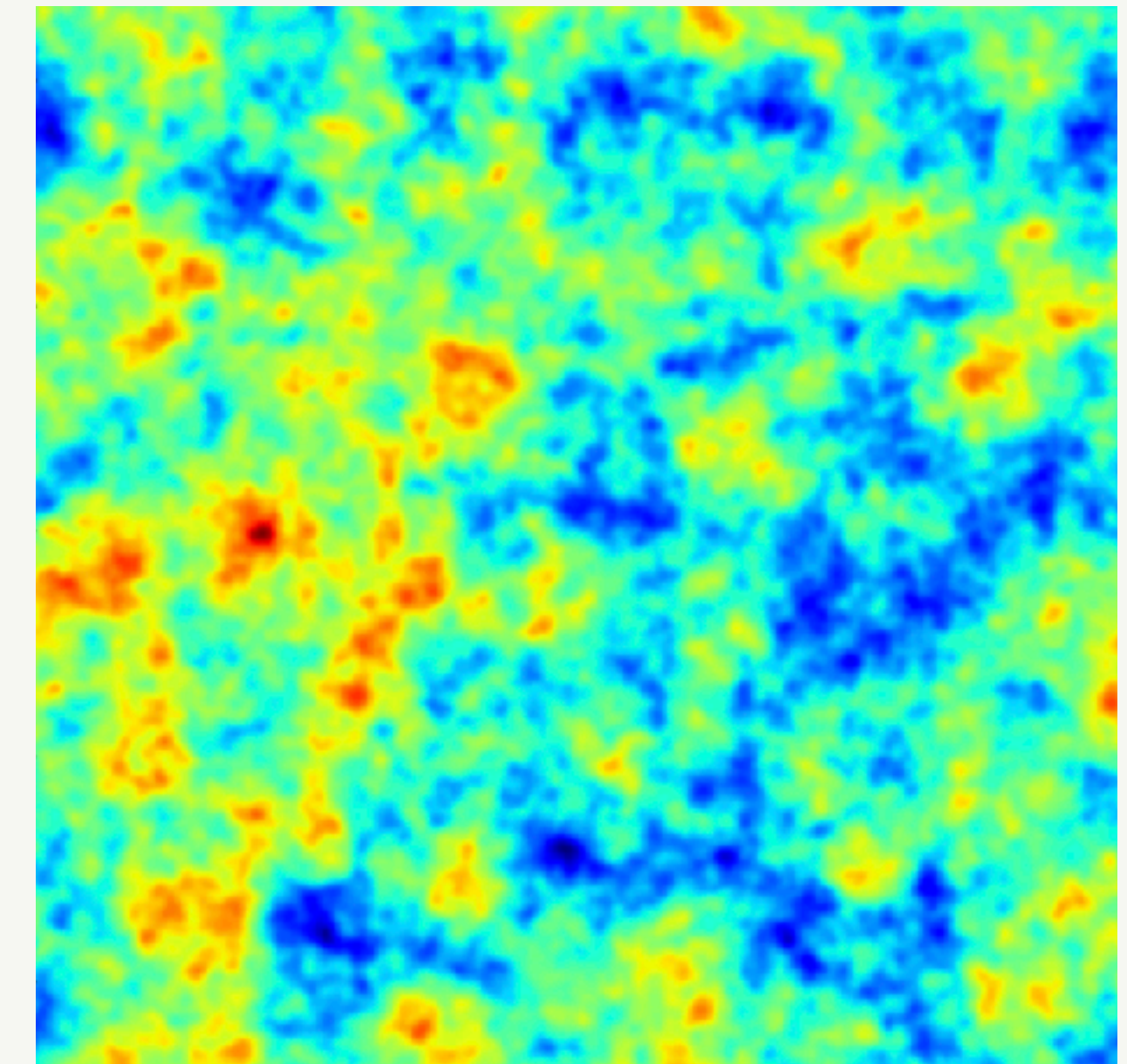
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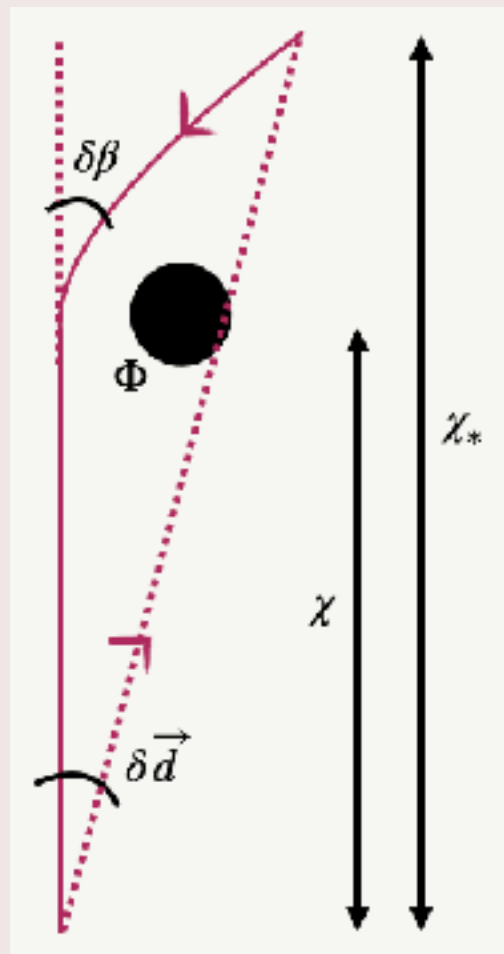


The lensed CMB temperature in a direction $\hat{n}$, $\tilde{T}(\hat{n})$, is given by the unlensed temperature in the deflected direction:

$$\tilde{T}(\hat{n}) = T(\hat{n}') = T(\hat{n} + \vec{d}), \quad \vec{d} = \nabla \phi$$

c.f. similar expressions can be written for the polarisation fields Q, U

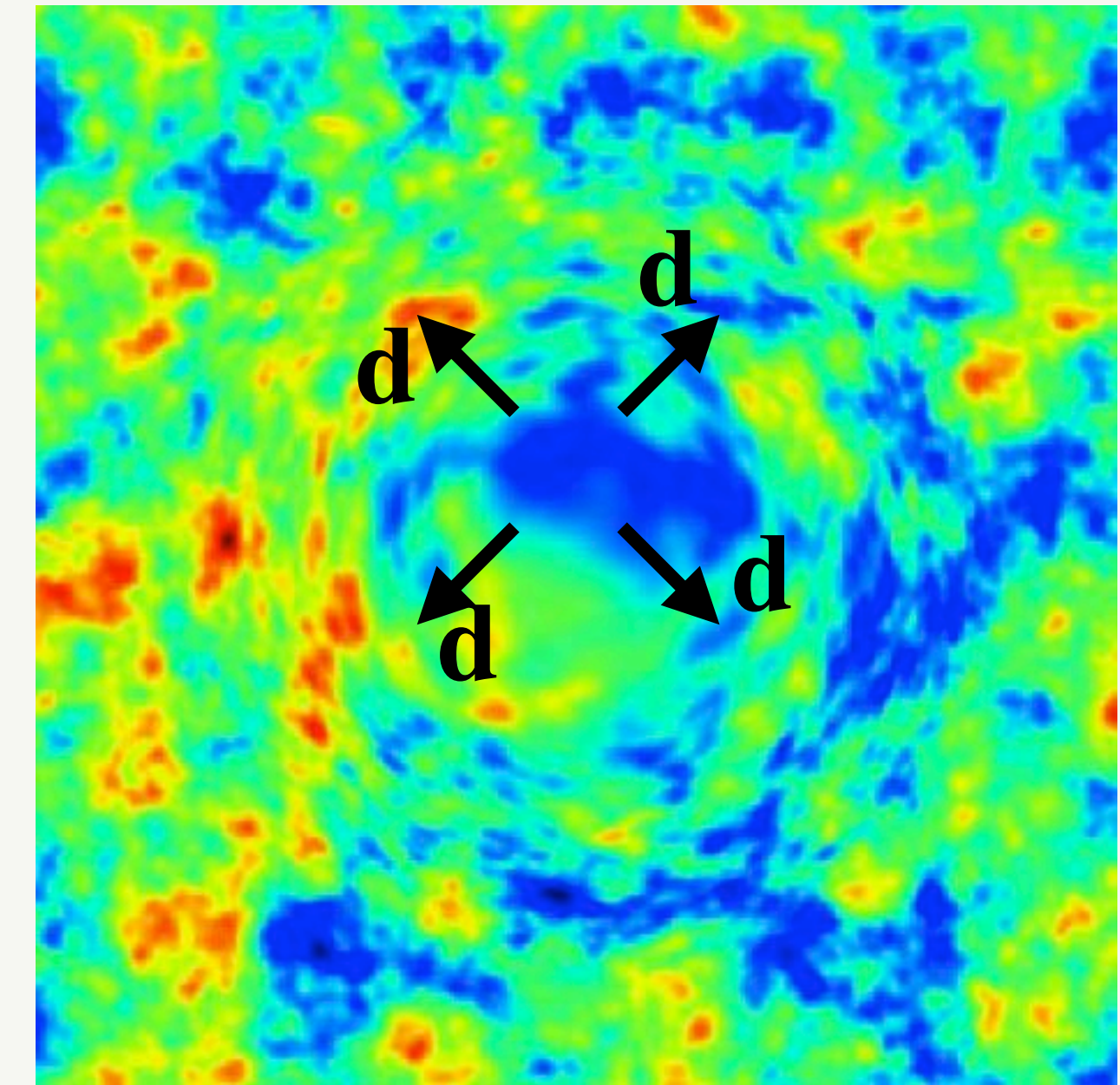


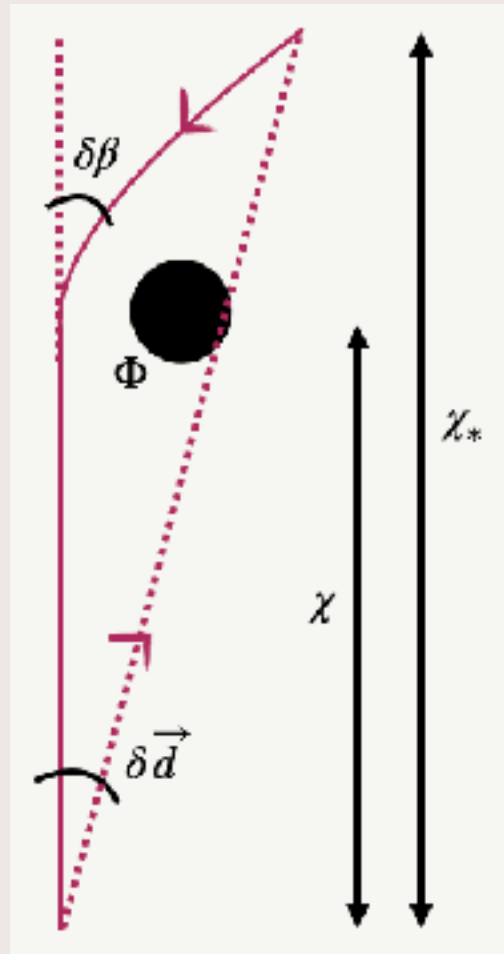


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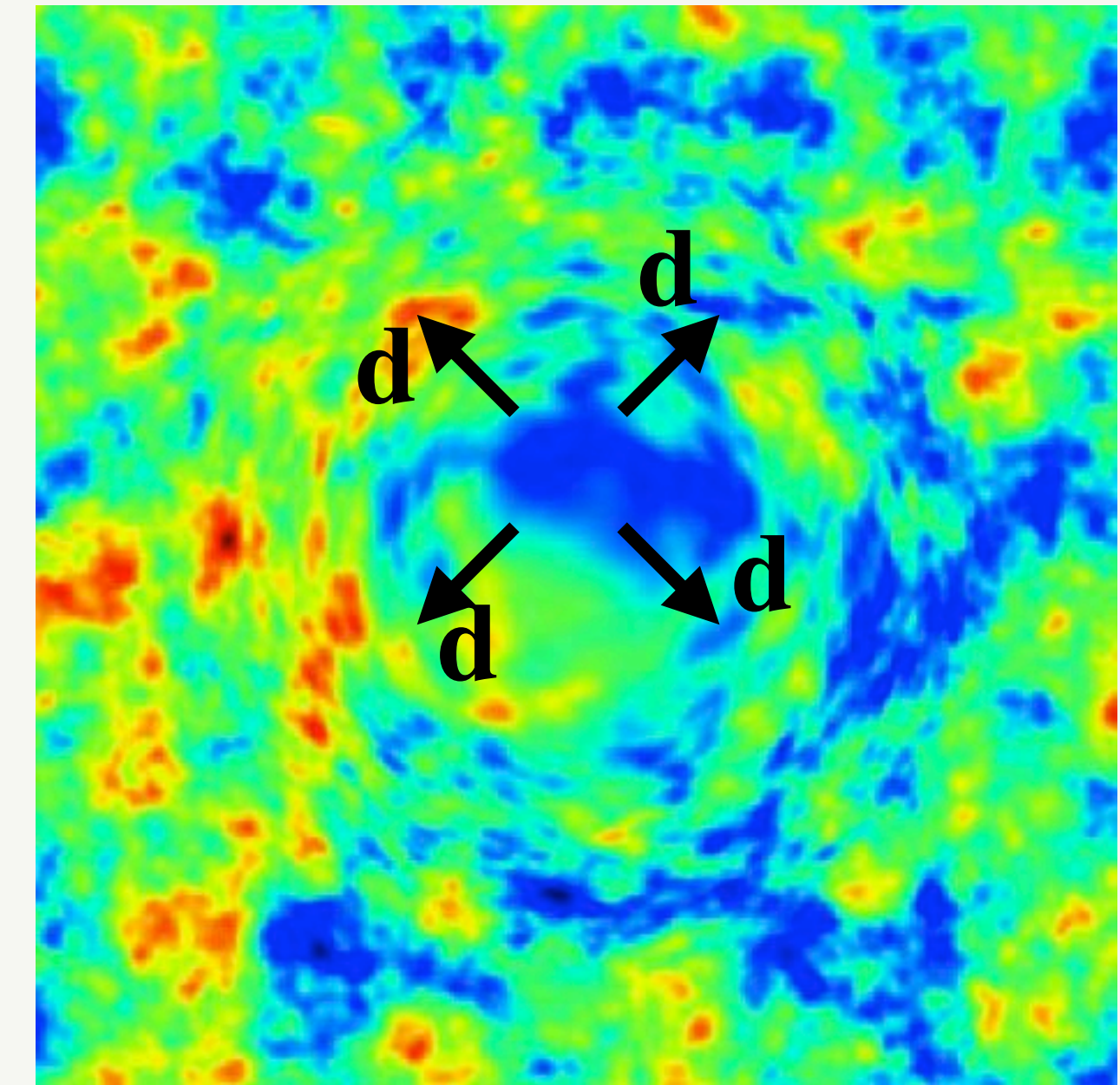




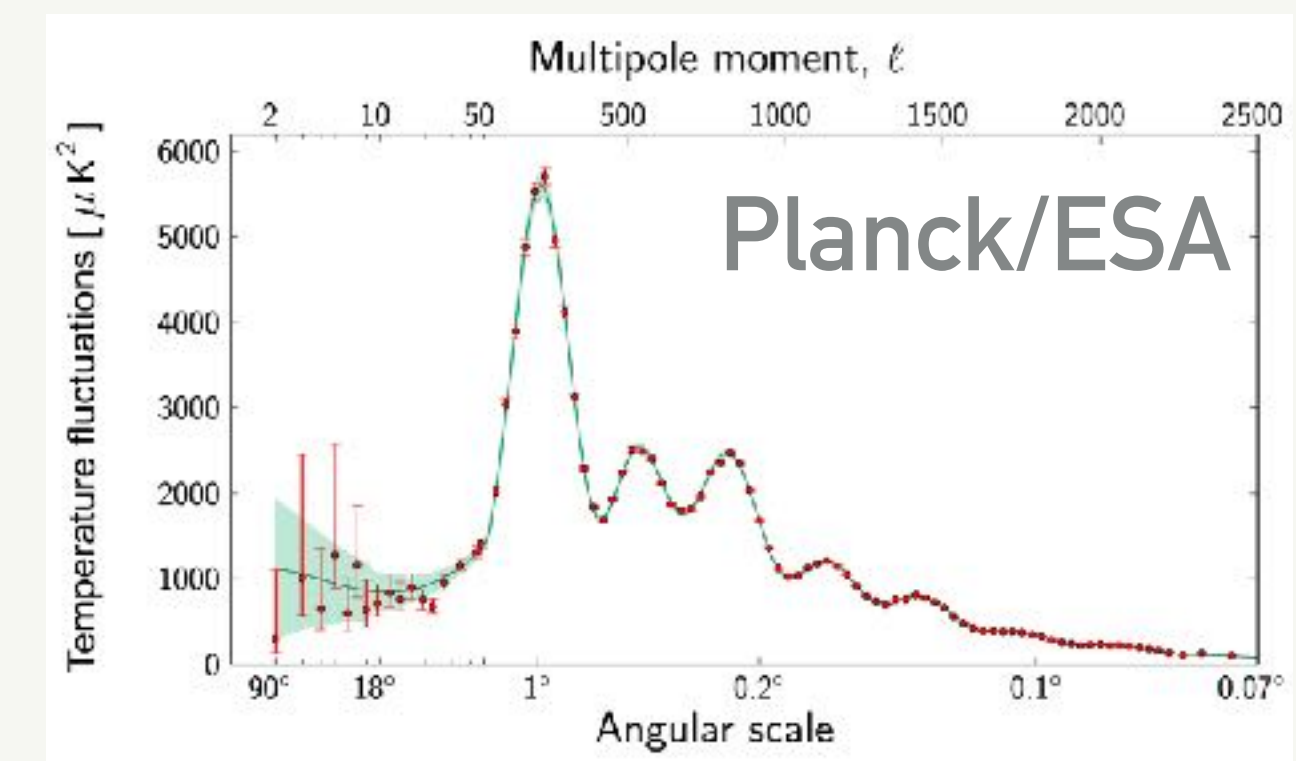
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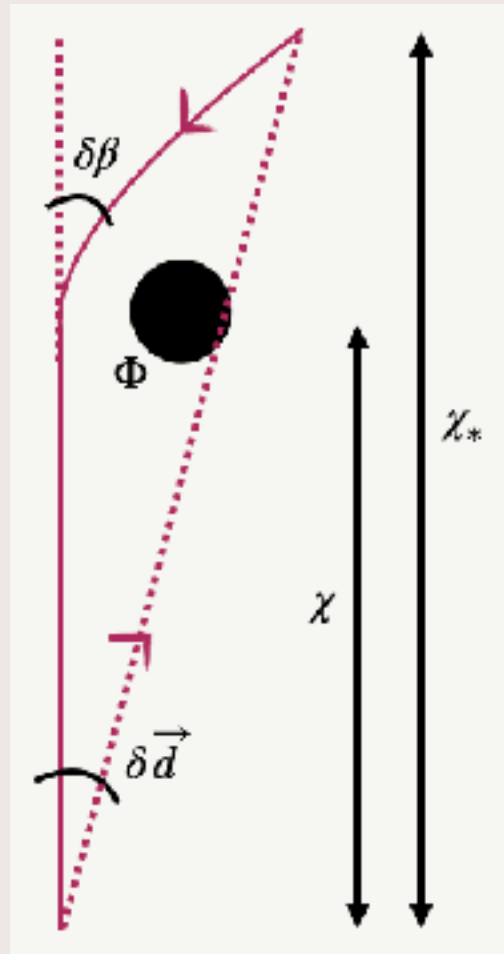
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Lensing effects on the CMB power spectra

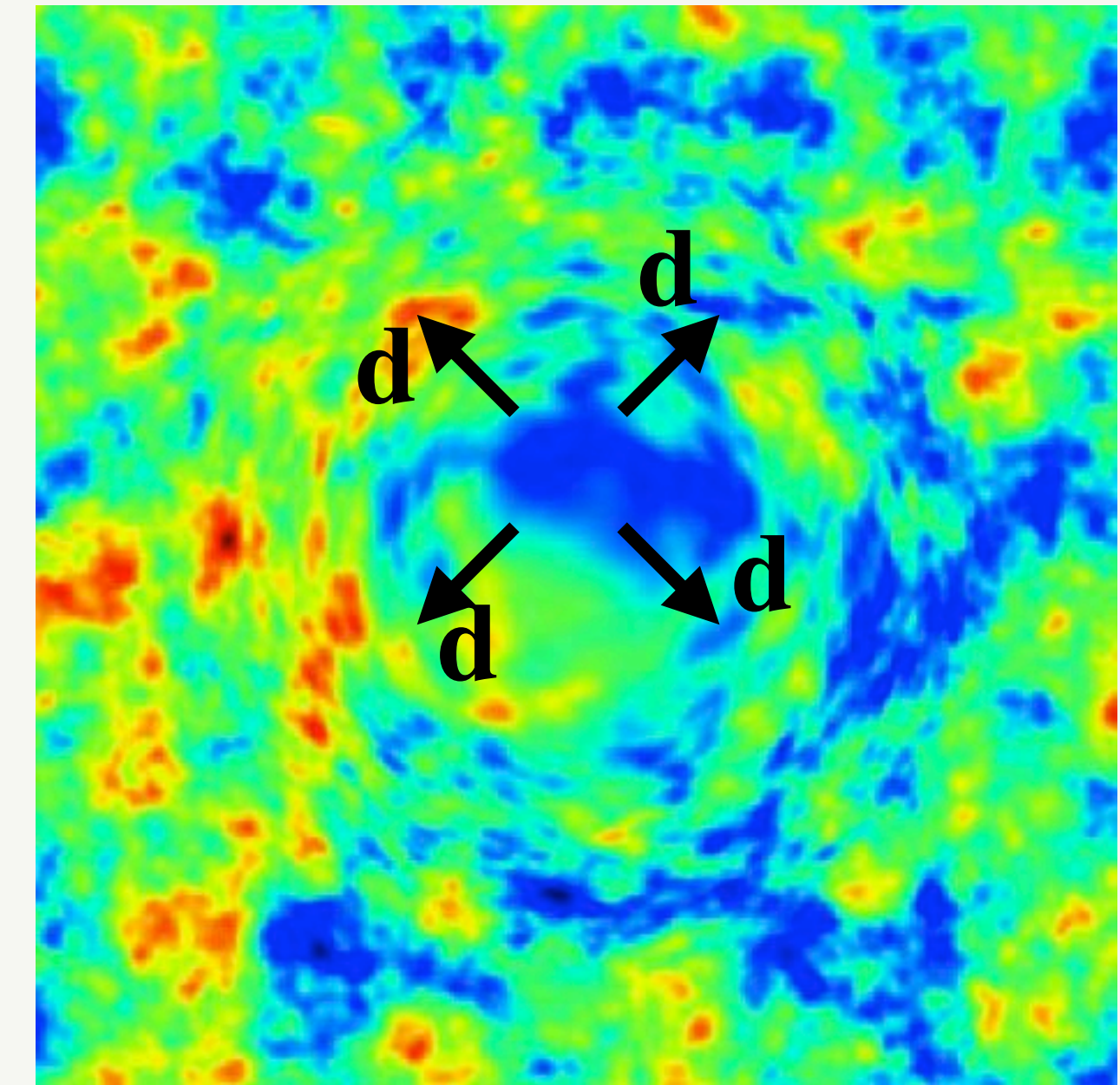




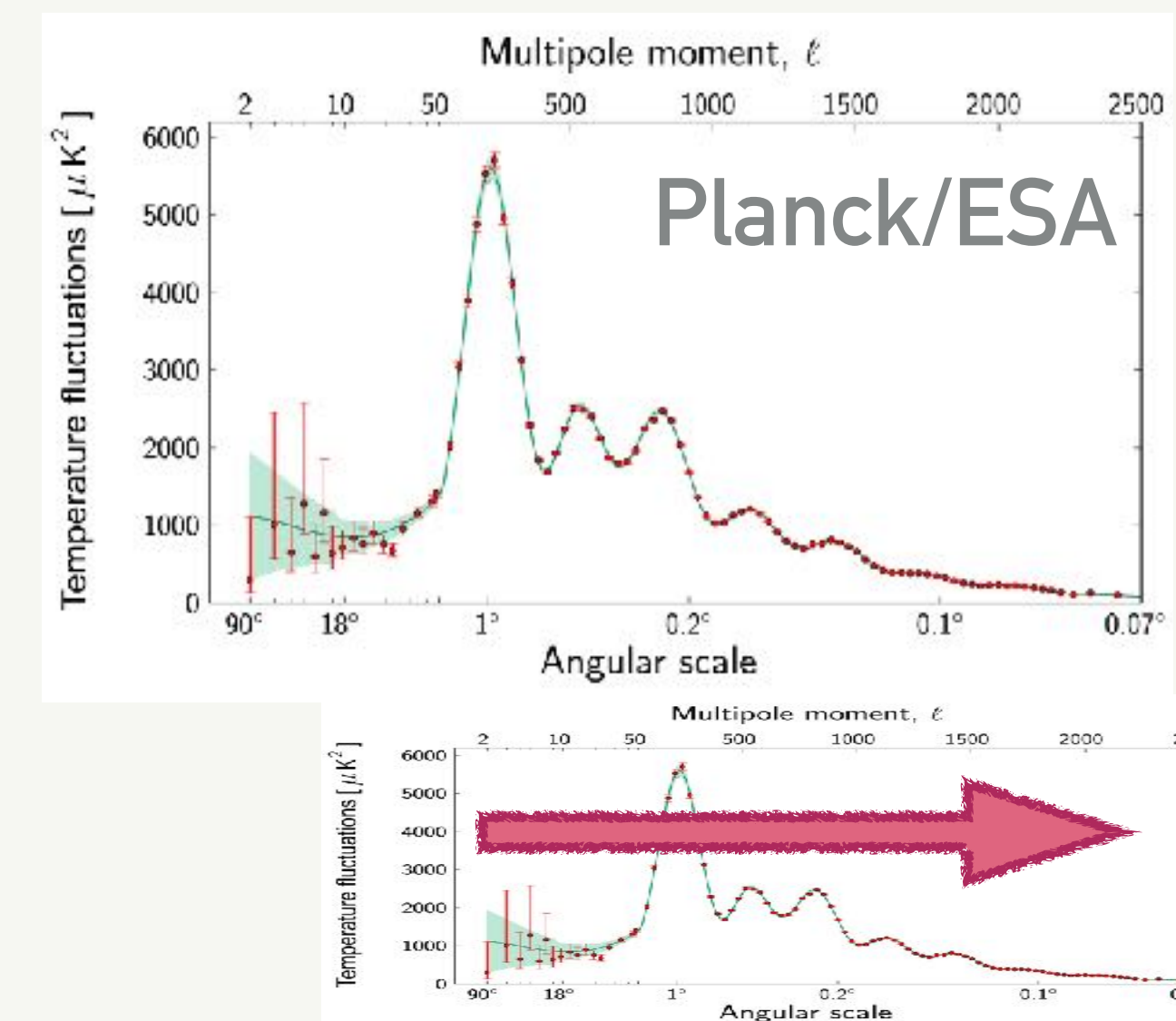
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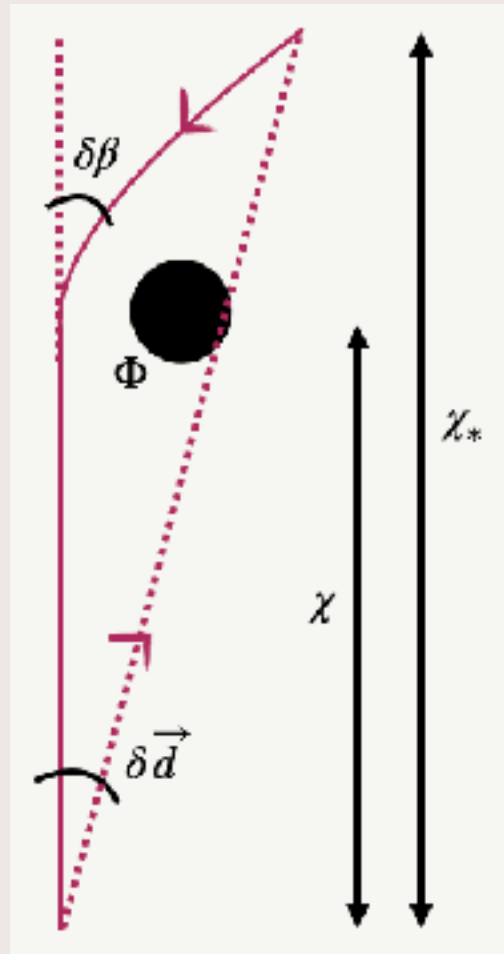
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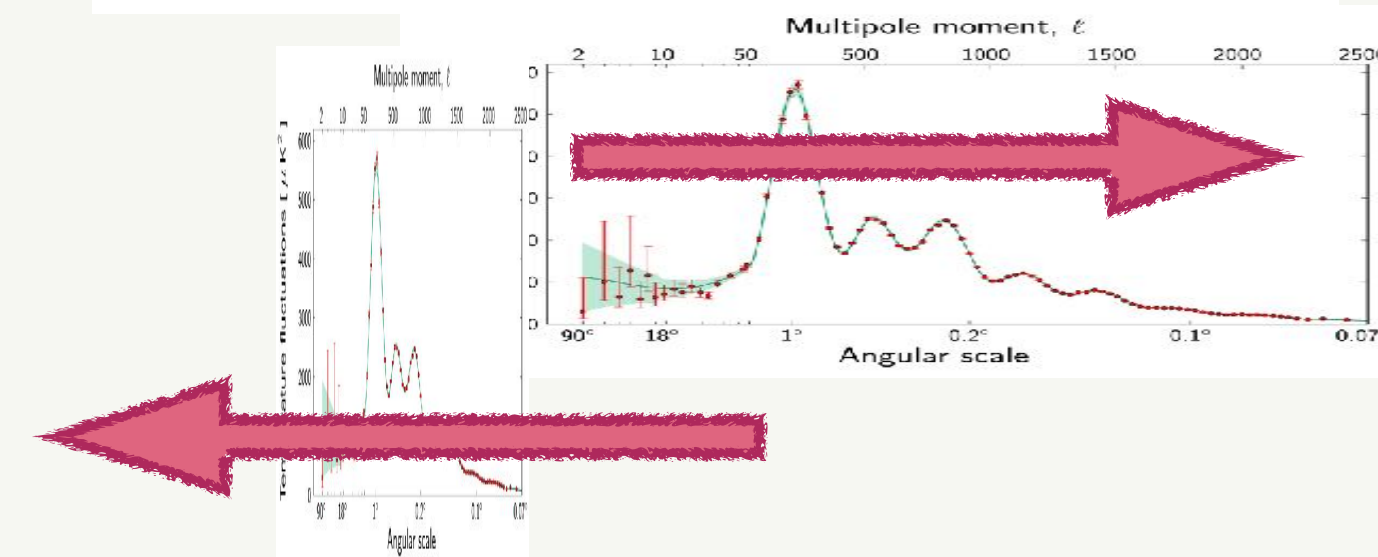
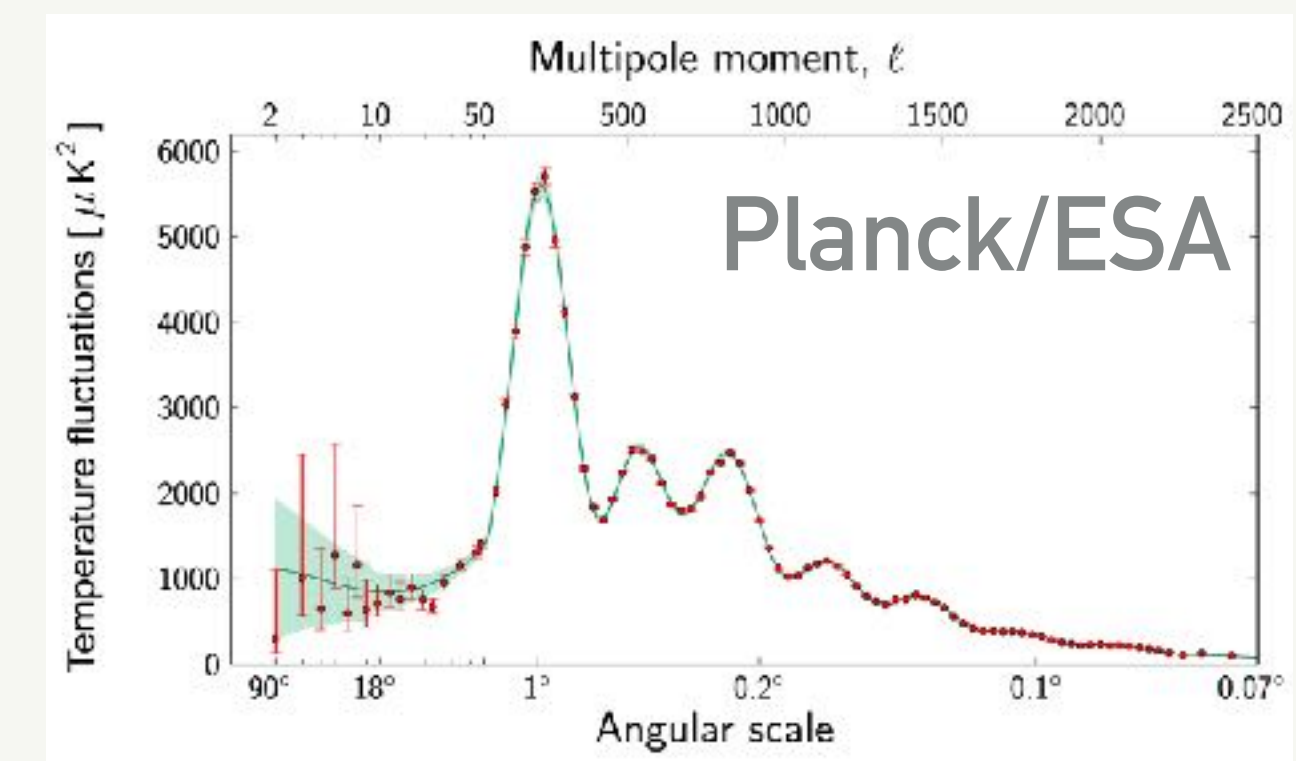
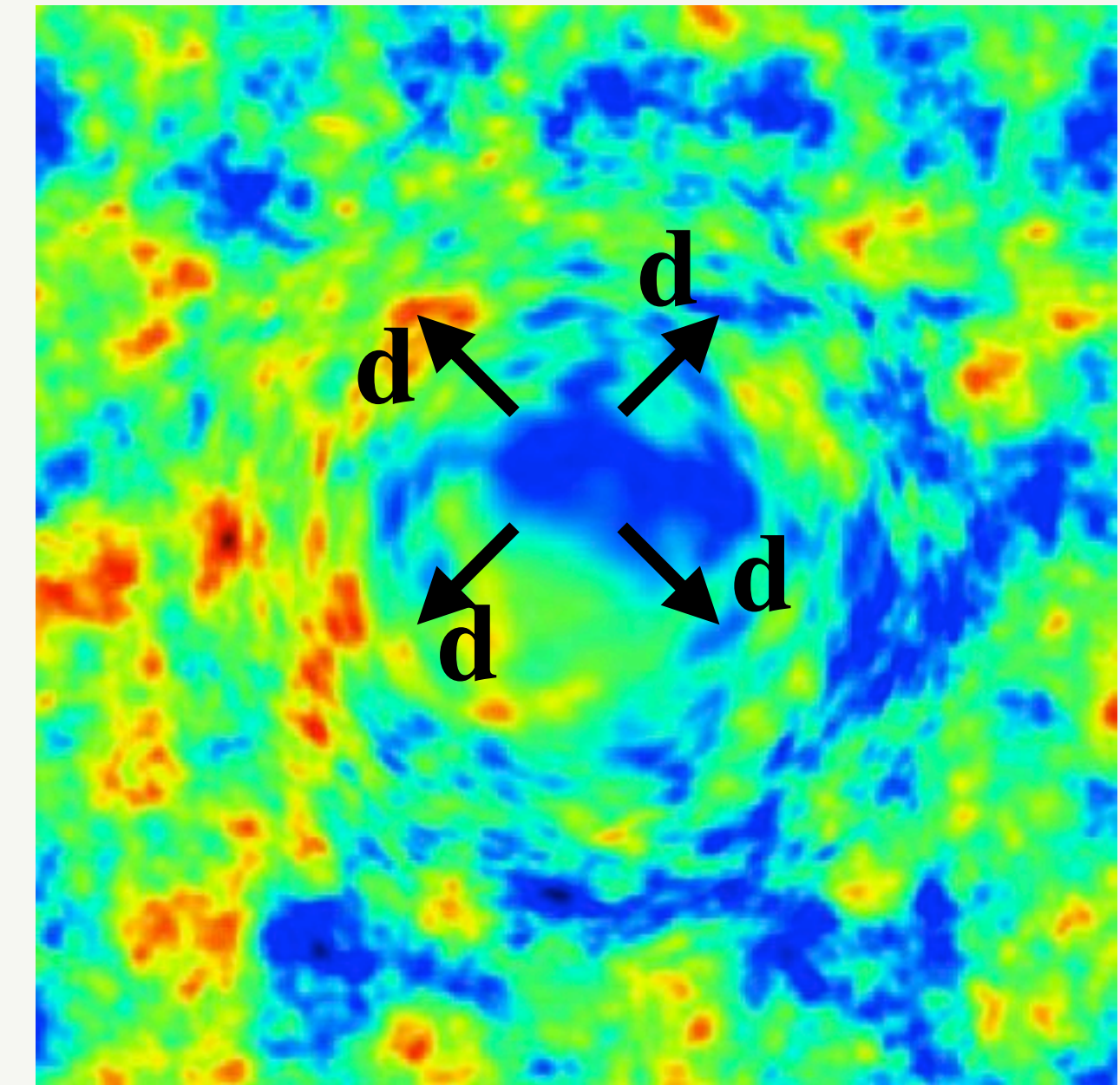


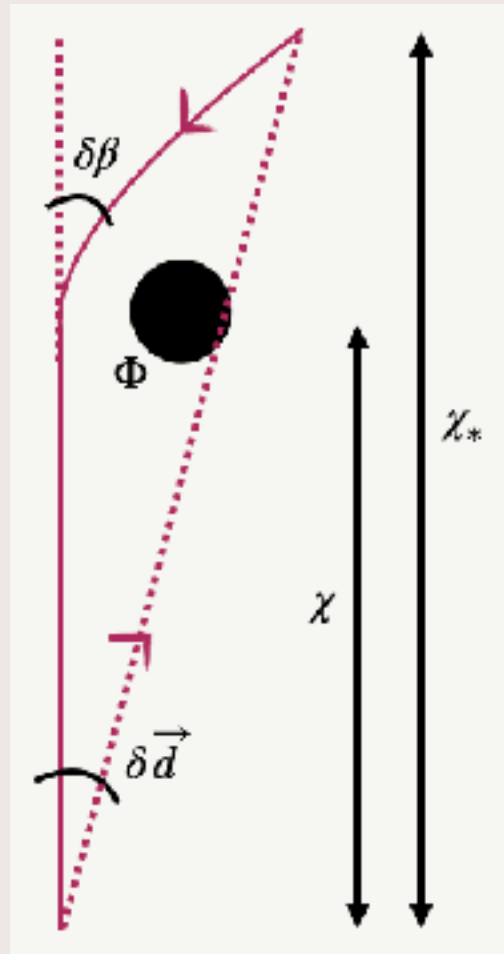
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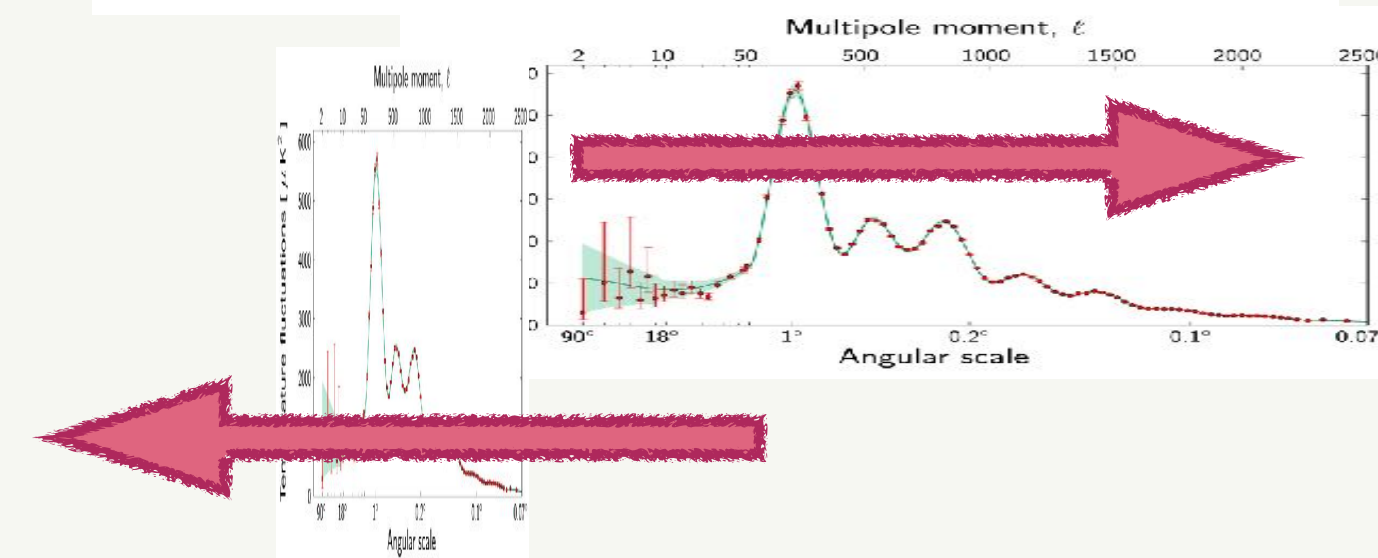
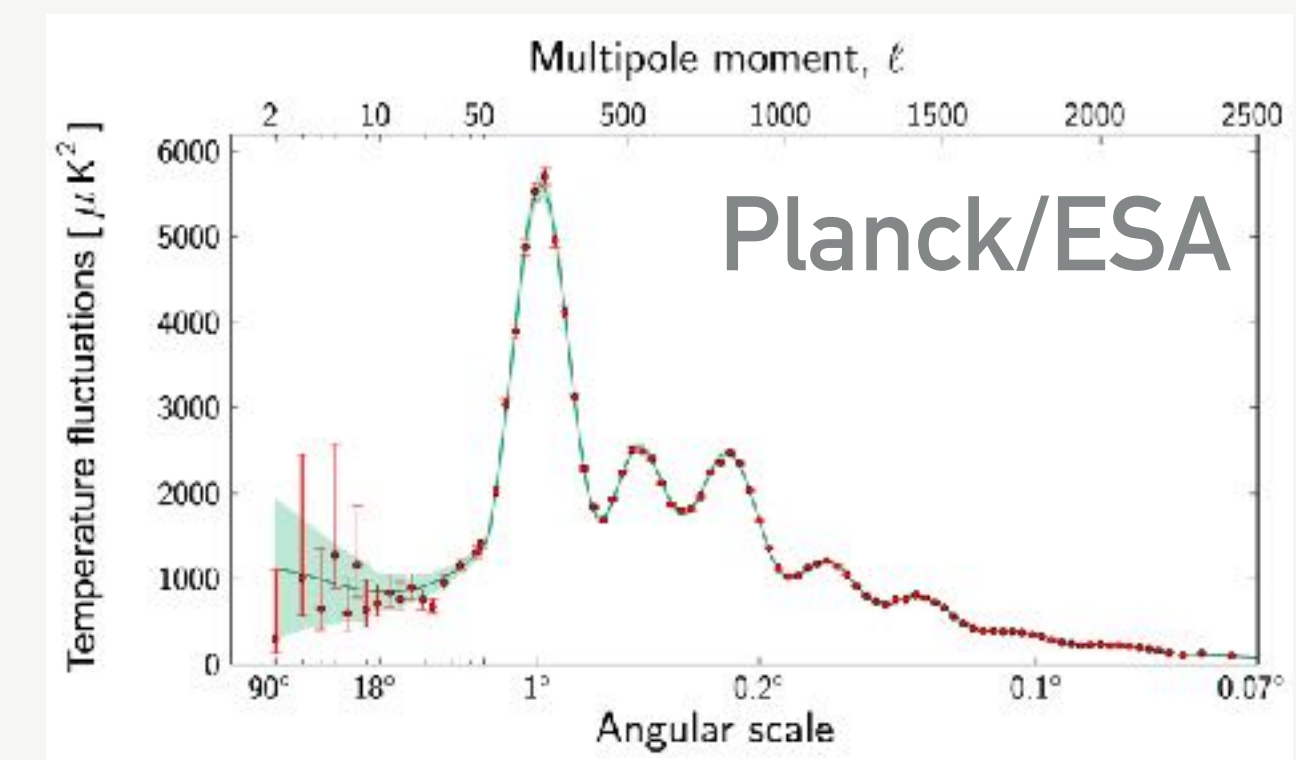
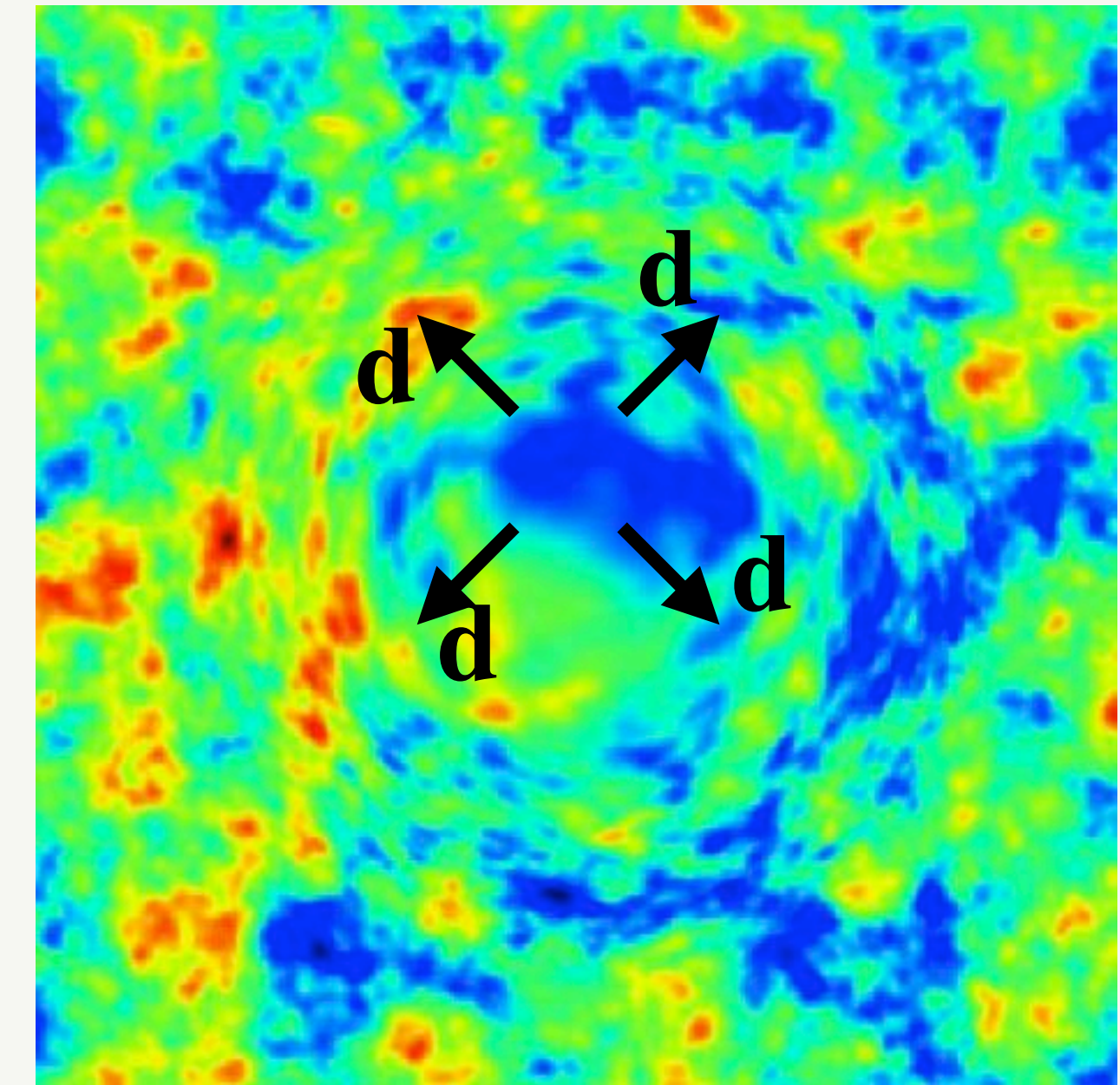
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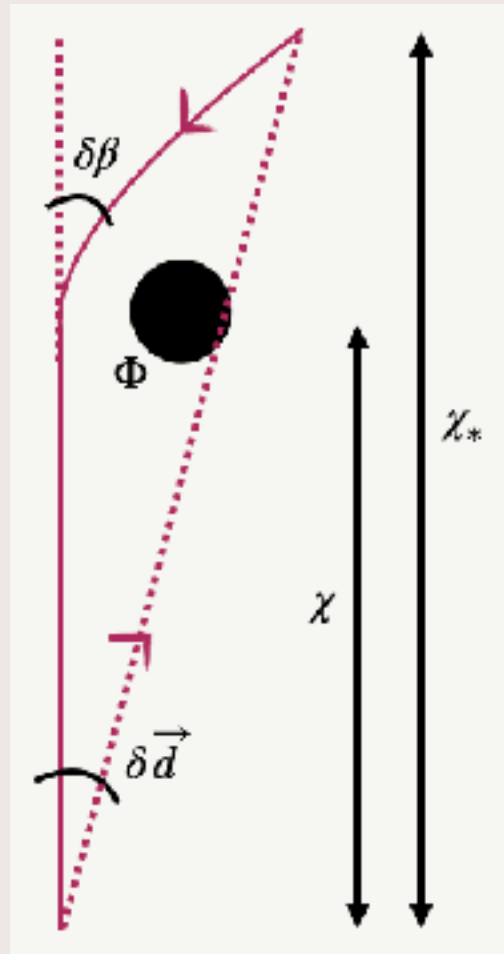
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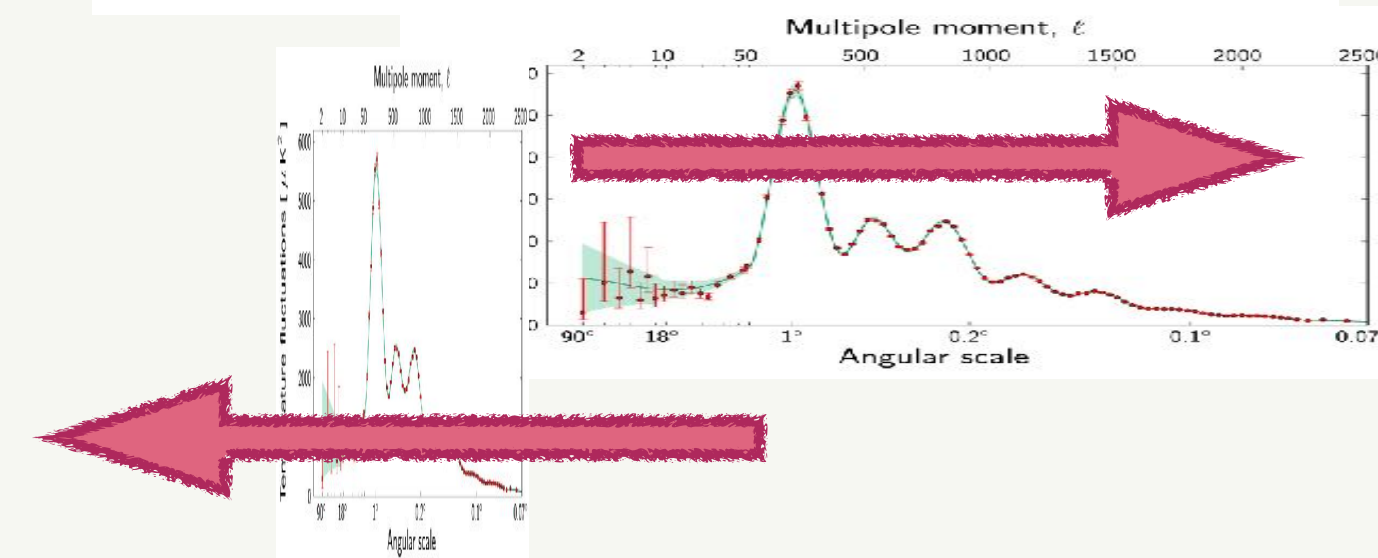
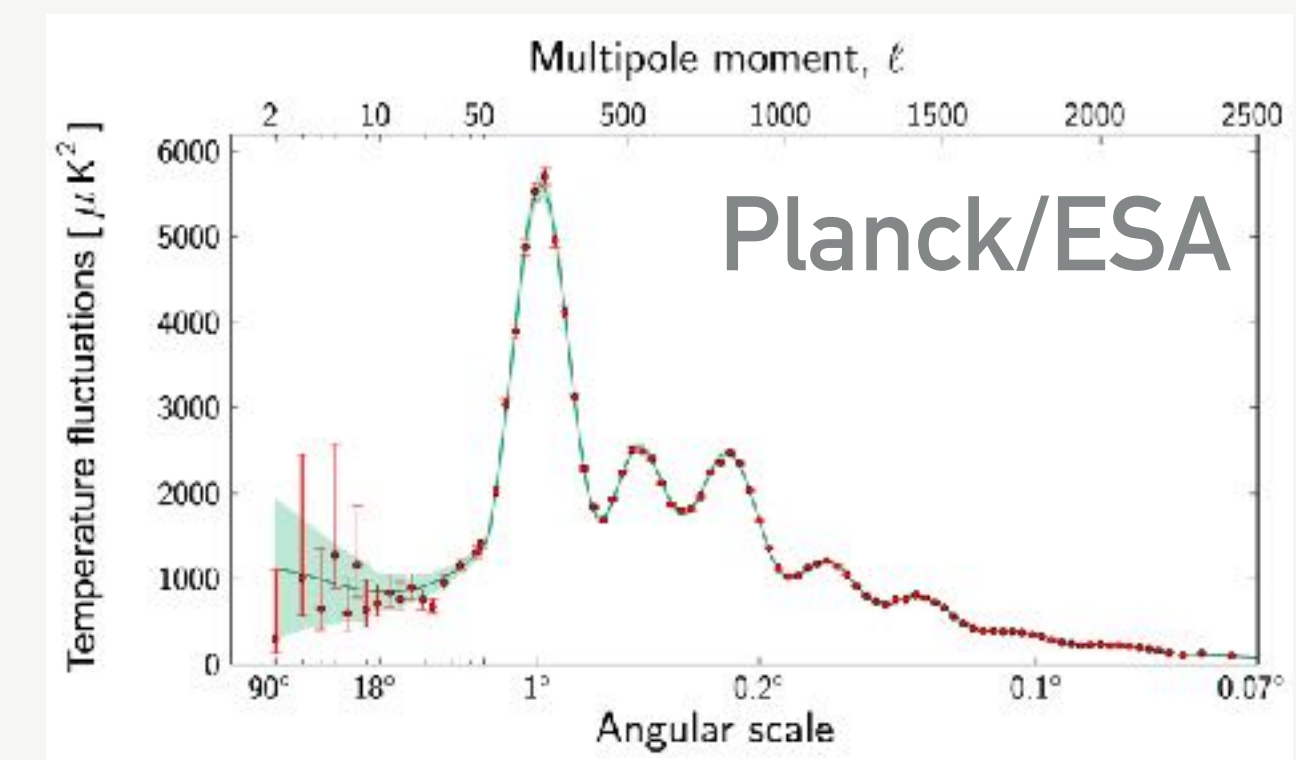
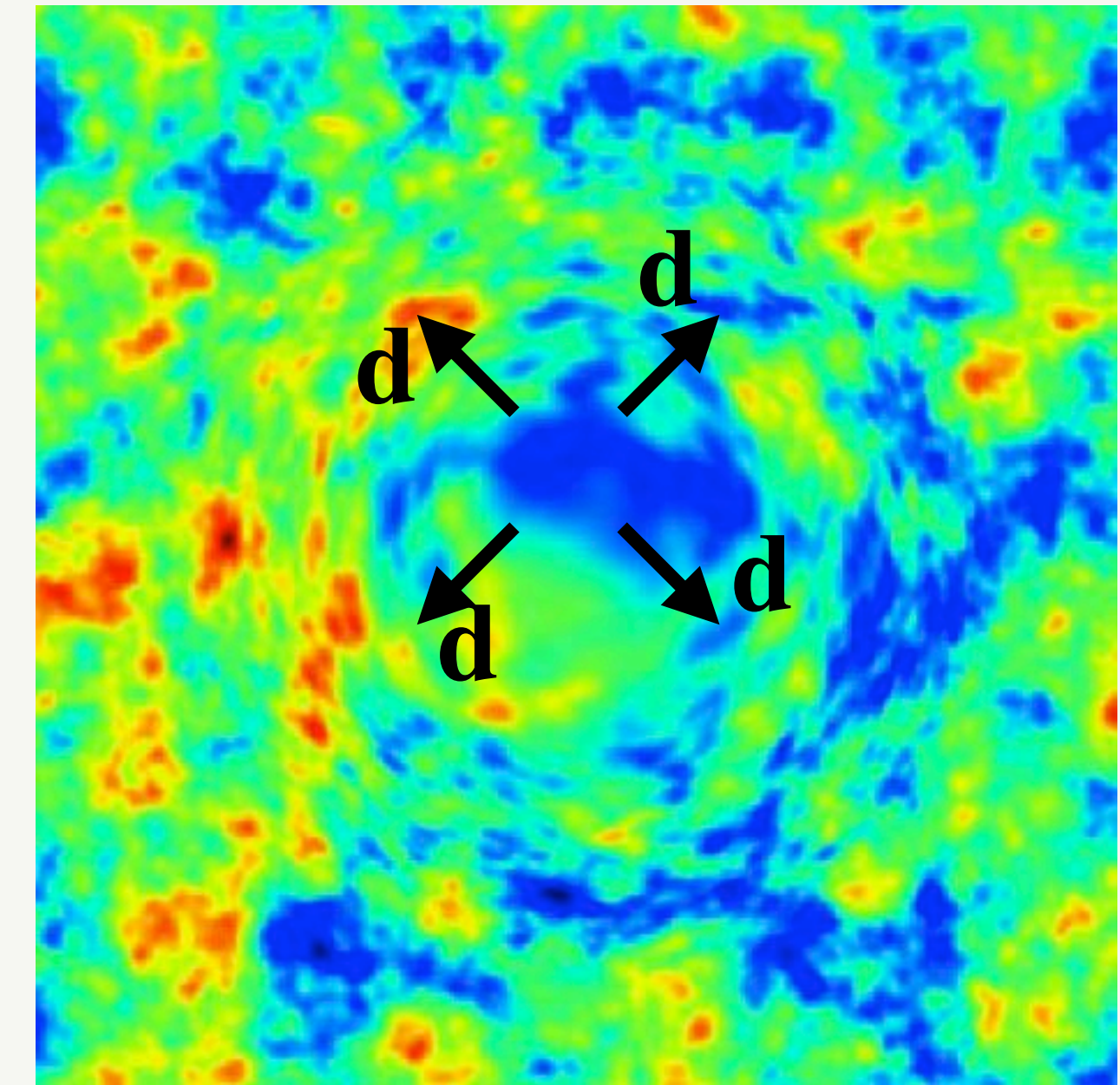
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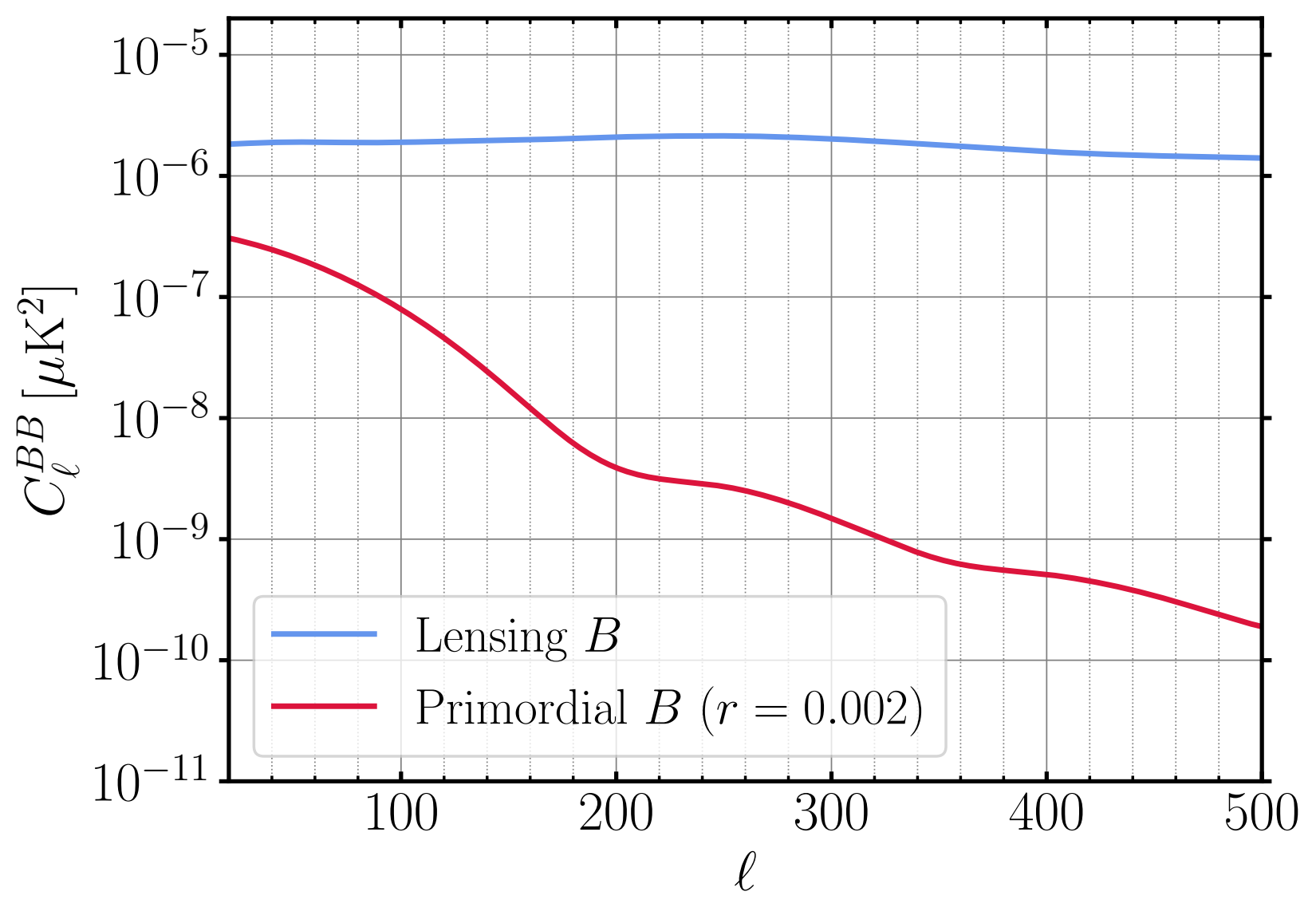
1. Large-scale lenses smooth the peaks of the observed power spectrum
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3. Lensing of the CMB polarisation causes power to leak from E- to B-modes (c.f. lensed B-modes are nearly white, $C_{\ell}^{BB} \sim 2 \times 10^{-6} \mu\text{K}^2$ corresponding to instrumental white-noise level of $5 \mu\text{K-arcmin}$)



For references, see e.g. Sherwin & Schmittfull

B-modes from the early Universe and delensing

“Answering the Big Questions”



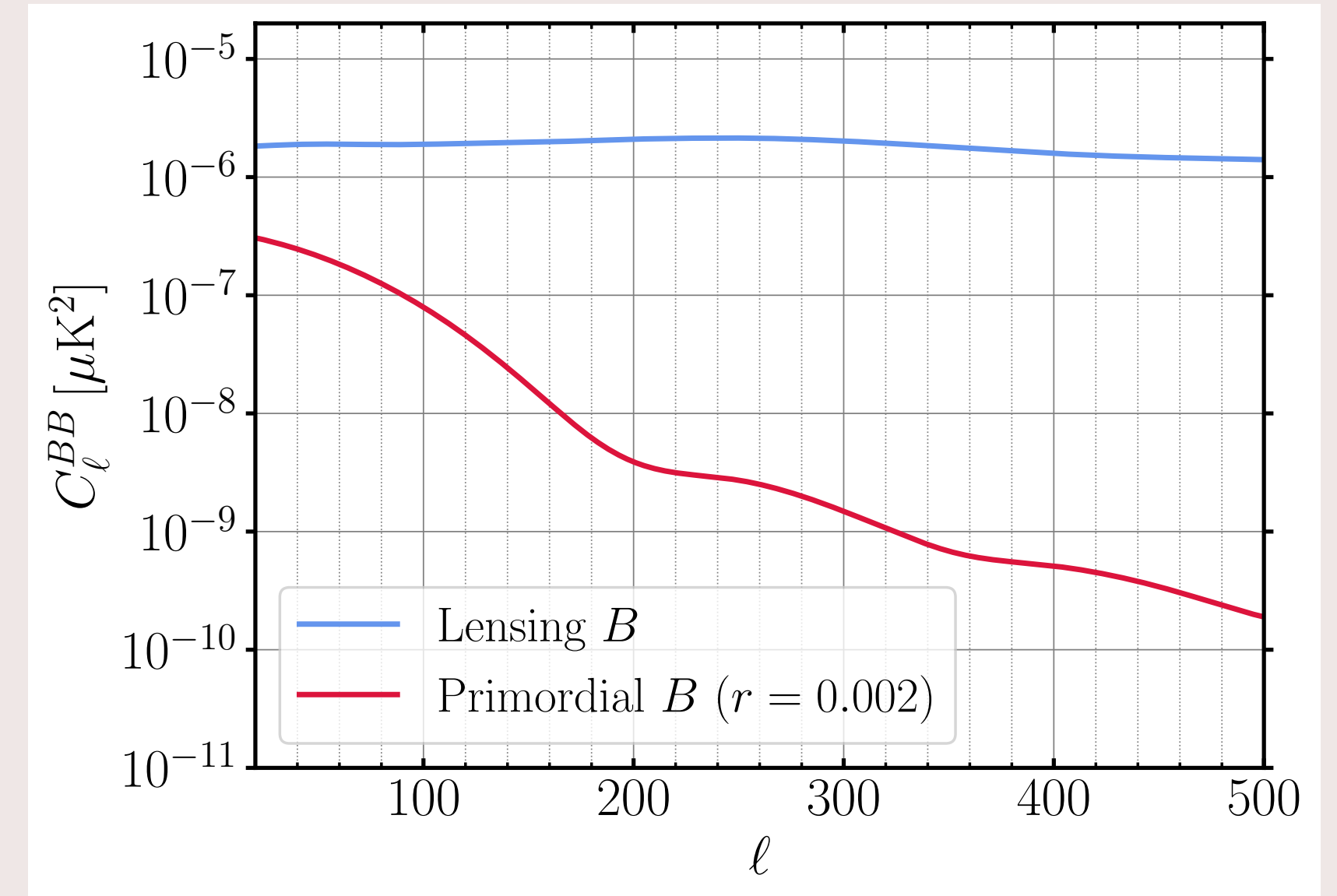
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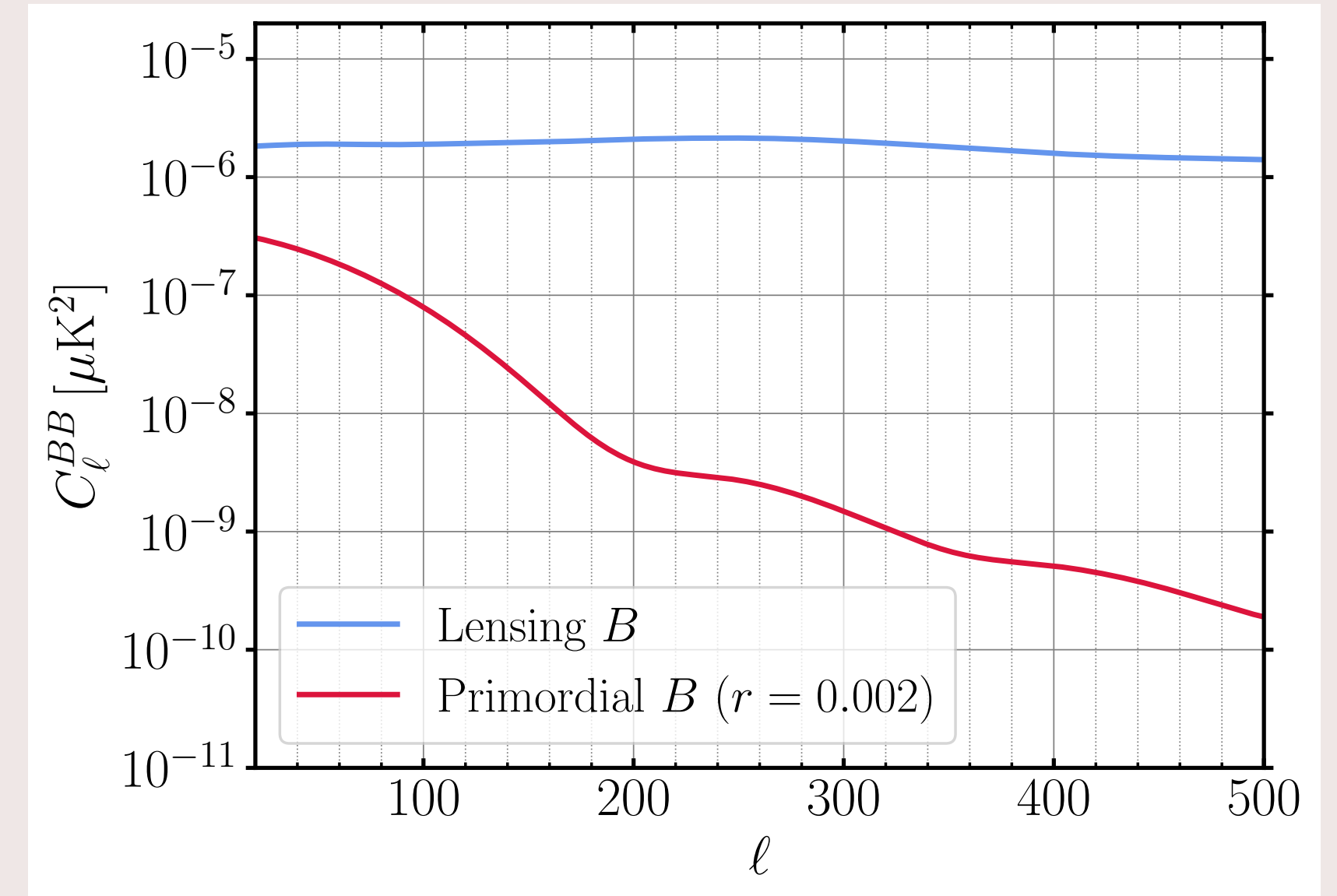
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$$\tilde{B}(\mathbf{l}) = \int \frac{d^2\mathbf{l}'}{(2\pi)^2} W(\mathbf{l}, \mathbf{l}') E(\mathbf{l}') \phi(\mathbf{l} - \mathbf{l}')$$



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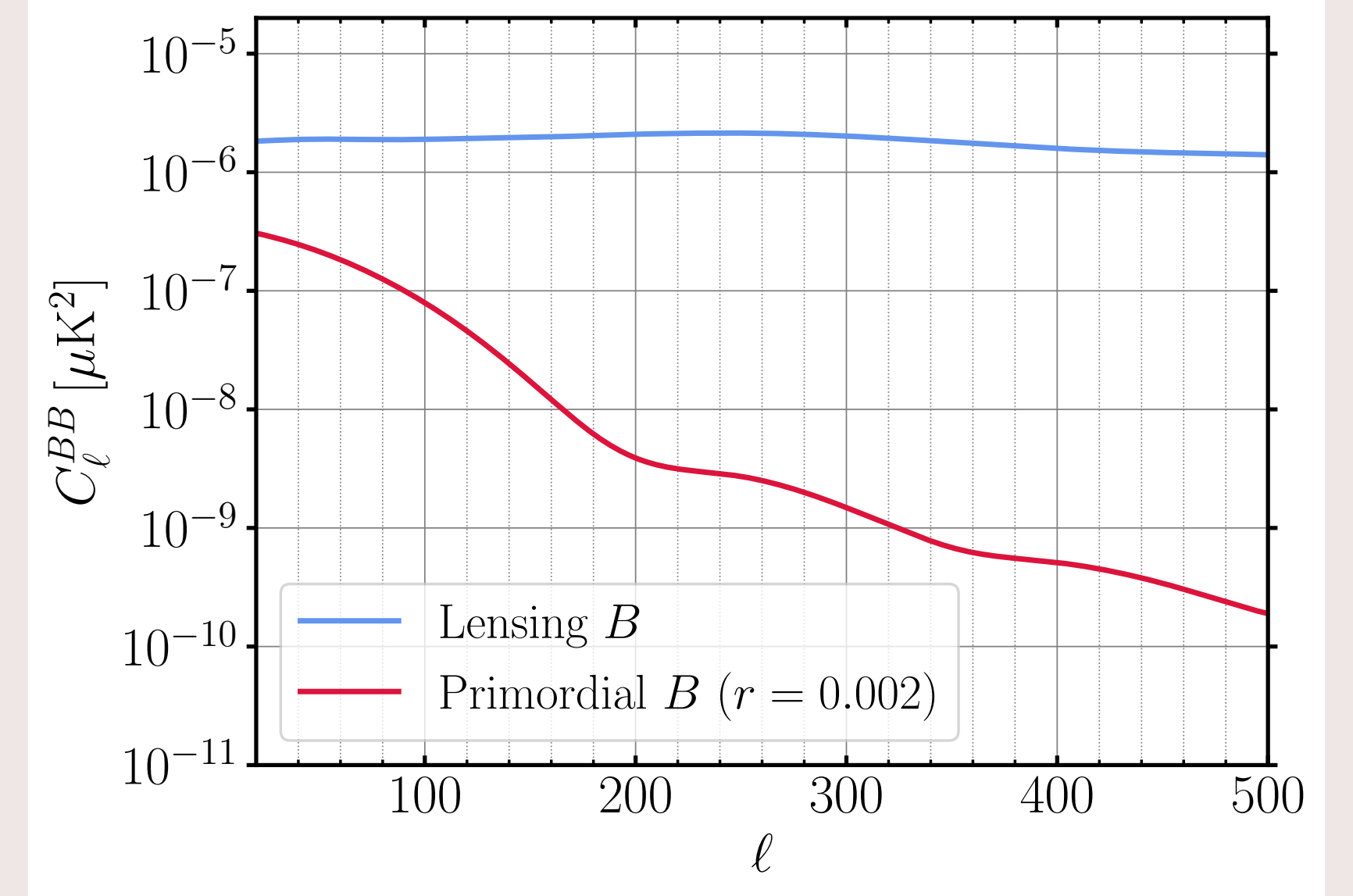
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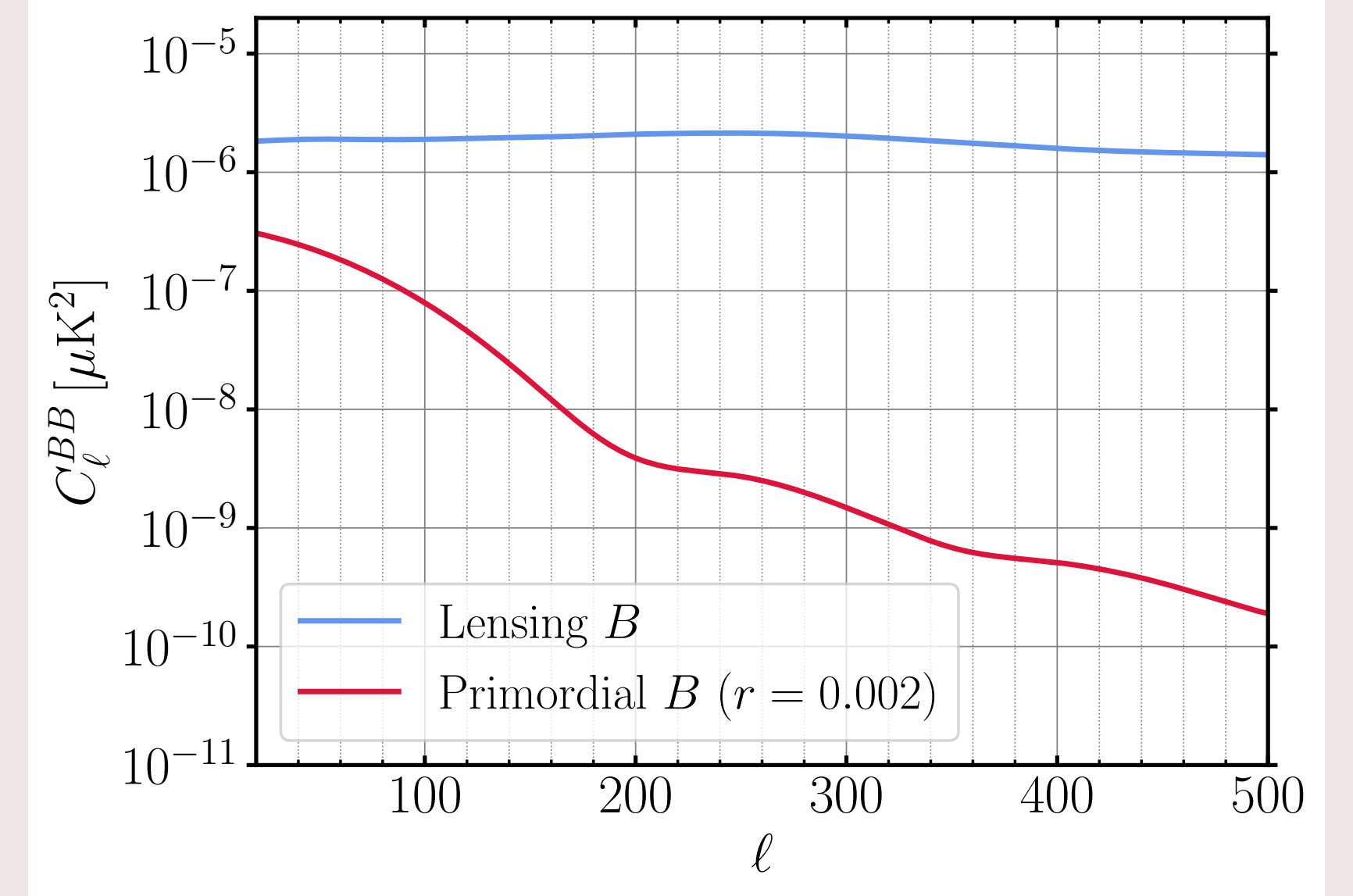
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- Want $f(\mathbf{l}, \mathbf{l}')$ to minimise residual B-mode power, and then:

$$C_l^{BB, \text{res}} = \int \frac{d^2 \mathbf{l}'}{(2\pi)^2} W^2(\mathbf{l}, \mathbf{l}') C_{l'}^{EE} C_{|\mathbf{l}-\mathbf{l}'|}^{\kappa\kappa} \left(1 - \rho_{|\mathbf{l}-\mathbf{l}'|}^2\right)$$



Correlation coefficient between the true lensing signal and the tracer I

$$\rho = \frac{C_l^{\kappa I}}{\sqrt{C_l^{\kappa\kappa} C_l^{II}}}$$

How to measure CMB lensing?

CMB lensing reconstruction

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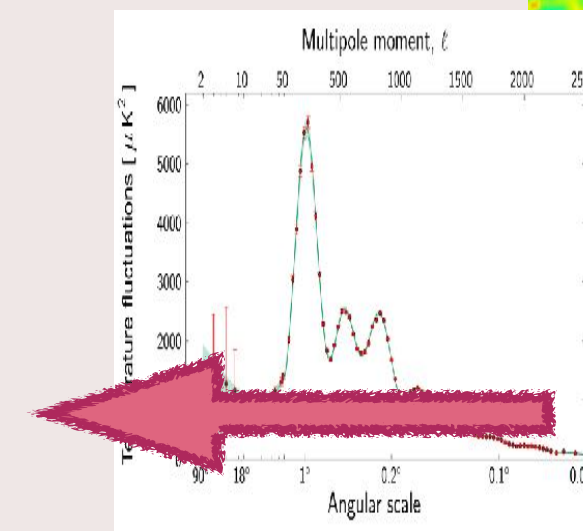
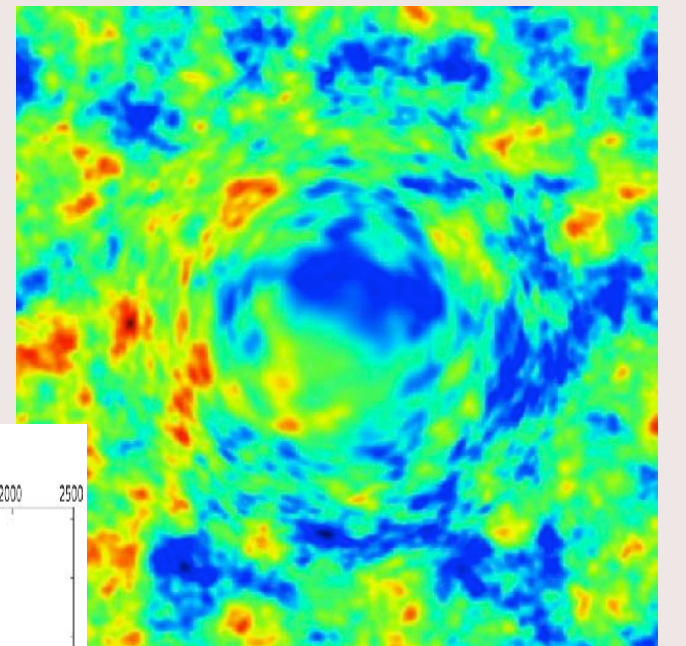
CMB lensing reconstruction

Statistical isotropy on CMB maps: $\langle T(\mathbf{l}) T(\mathbf{l}') \rangle = (2\pi)^2 C_l \delta^{(D)}(\mathbf{l} - \mathbf{l}')$

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shift to larger
angular scales

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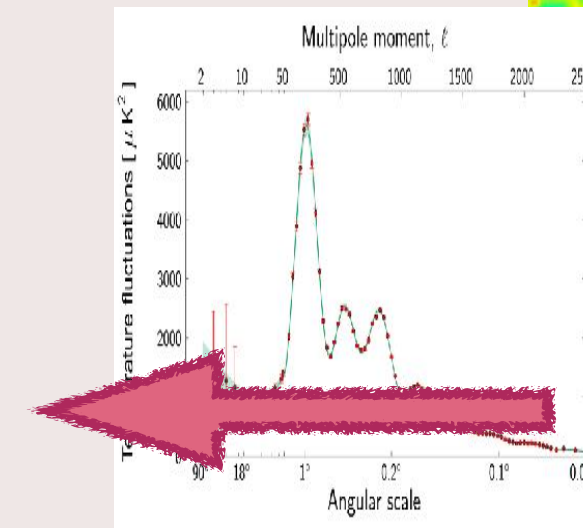
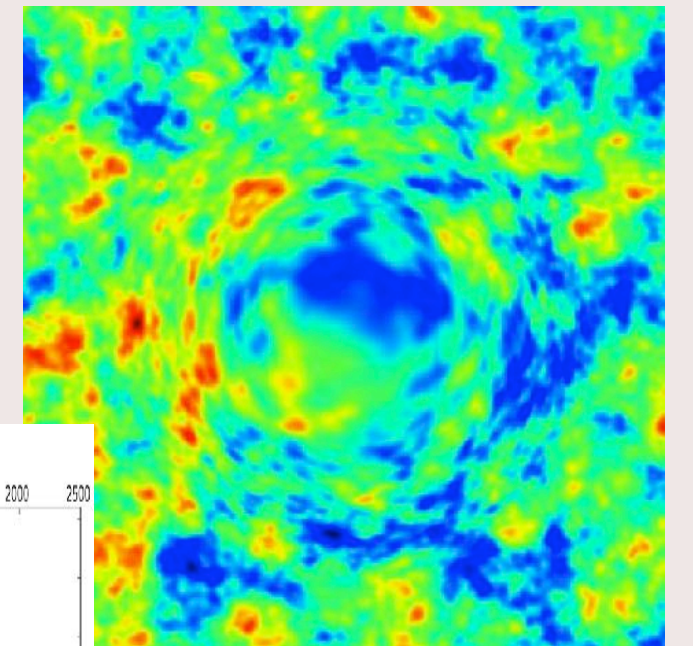
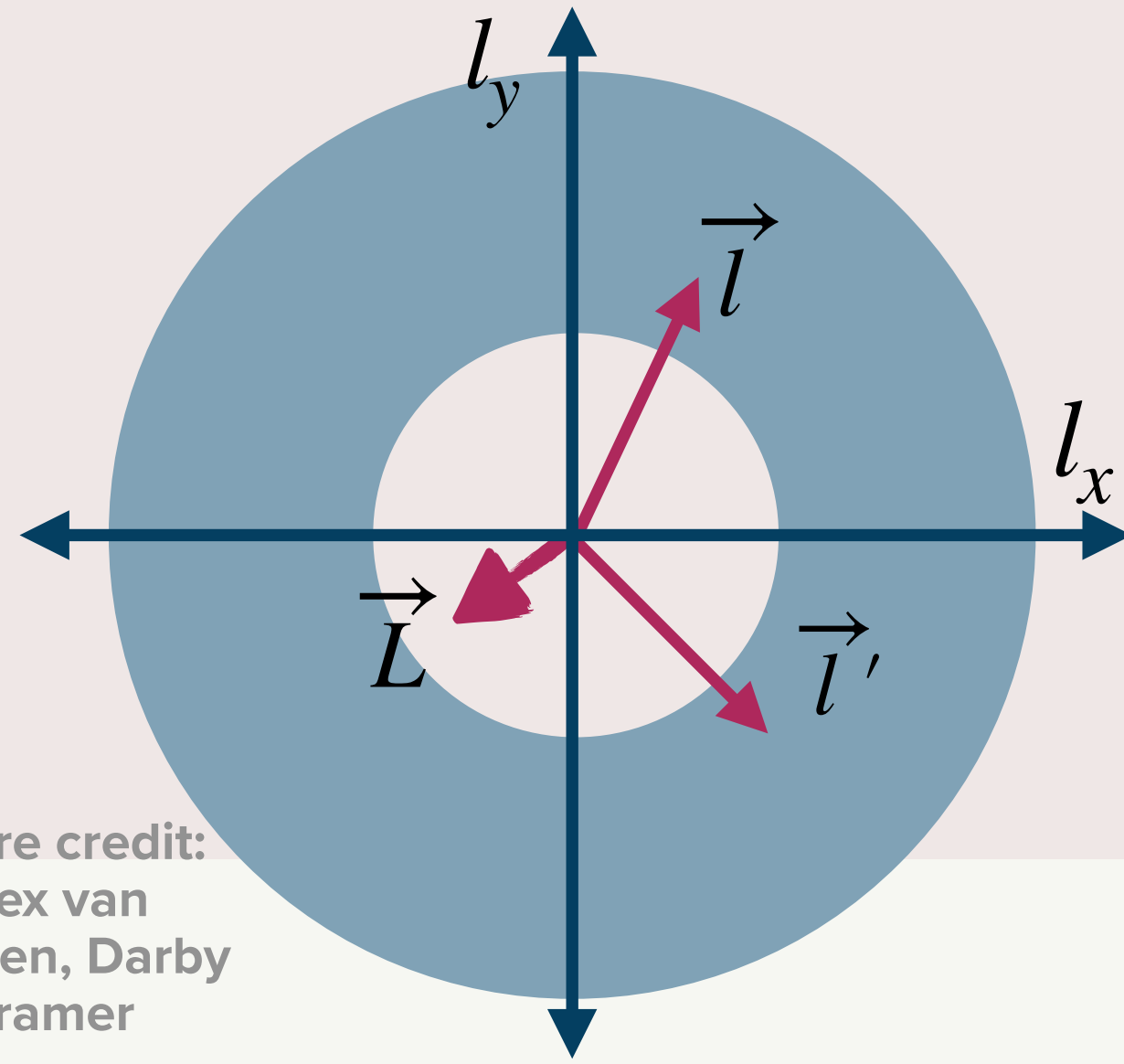
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with $\mathbf{L} = \mathbf{l}' - \mathbf{l}$.



shift to larger
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Recall:

$$\tilde{T}(\mathbf{l}) = T(\mathbf{l}) - \int \frac{d\mathbf{l}'}{(2\pi)^2} [\mathbf{l}' \cdot \mathbf{T}(\mathbf{l}')] \cdot (\mathbf{l} - \mathbf{l}') \phi(\mathbf{l} - \mathbf{l}')$$

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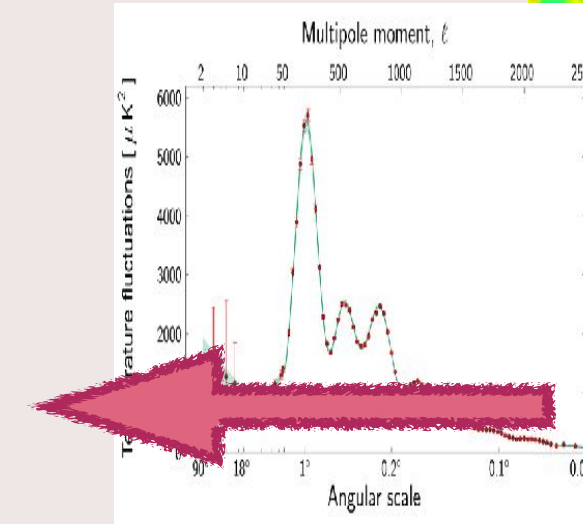
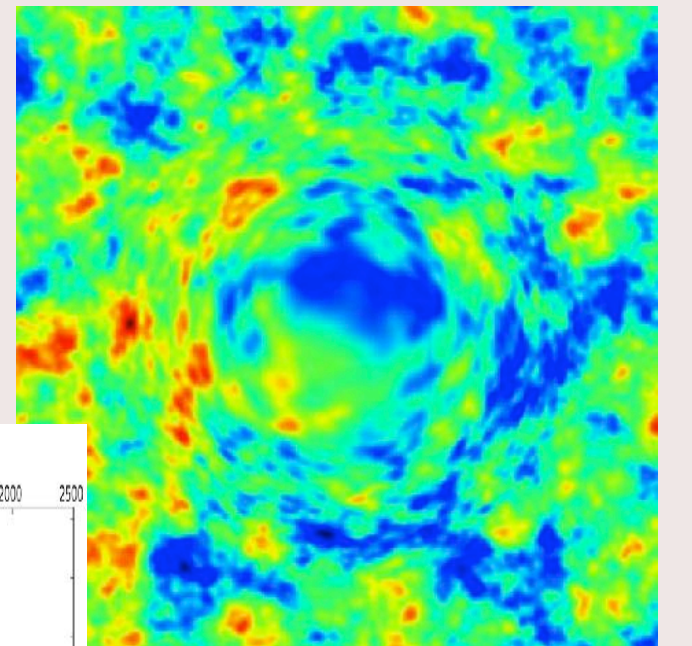
Recall:

$$\tilde{T}(\mathbf{l}) = T(\mathbf{l}) - \int \frac{d\mathbf{l}'}{(2\pi)^2} [\mathbf{l}' \cdot \mathbf{T}(\mathbf{l}')] \cdot (\mathbf{l} - \mathbf{l}') \phi(\mathbf{l} - \mathbf{l}')$$

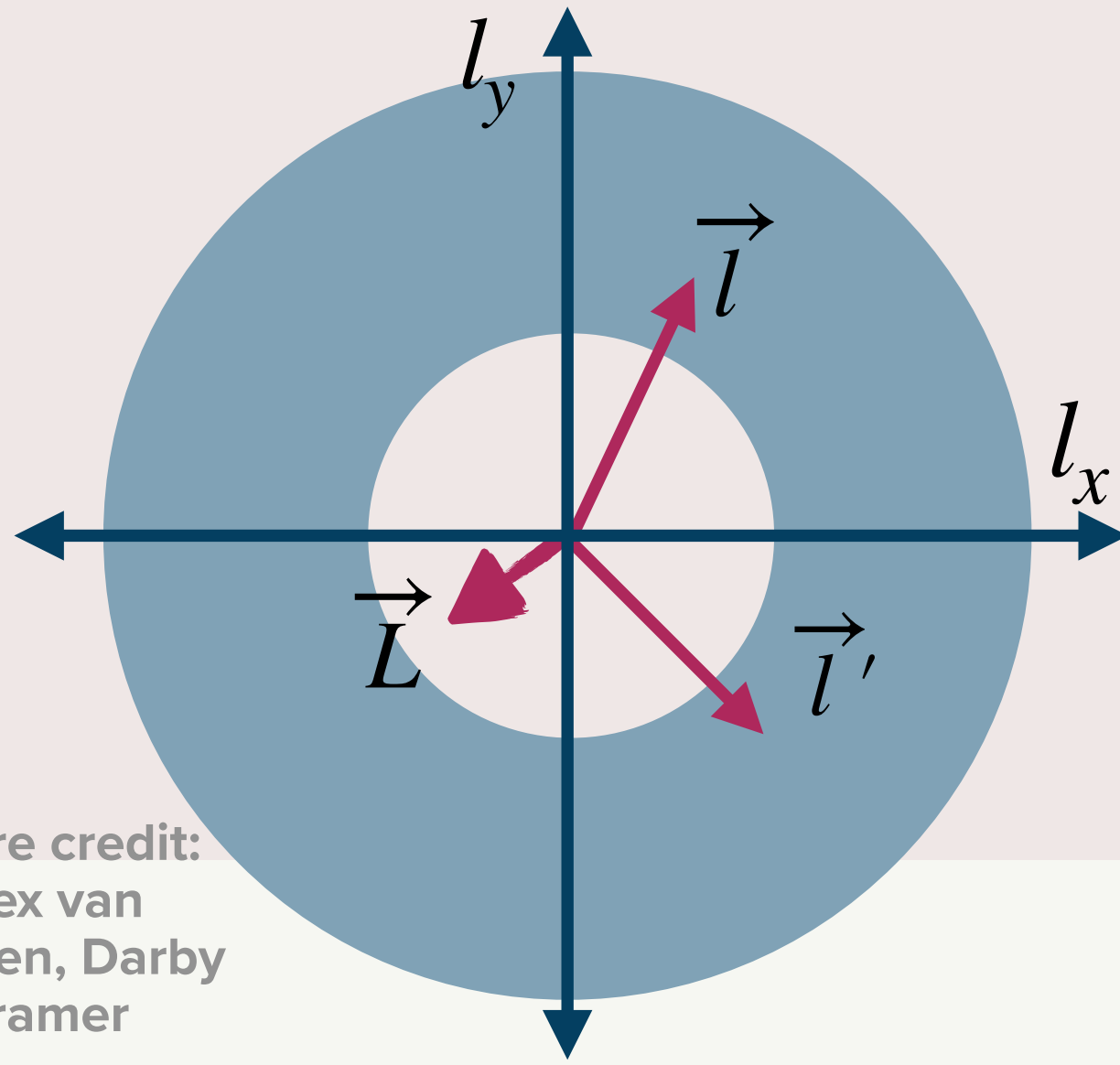
$$\hat{\phi} \sim \frac{\tilde{T}(\mathbf{l}) \tilde{T}(\mathbf{L} - \mathbf{l})}{K(\mathbf{l}, \mathbf{L})}$$

Combine all pairs with inverse-variance weighting:

$$\hat{\phi} \sim \int \frac{d^2\mathbf{l}}{(2\pi)^2} \frac{1}{\text{var} [\tilde{T}(\mathbf{l}) \tilde{T}(\mathbf{L} - \mathbf{l}) / K(\mathbf{l}, \mathbf{L})]} \frac{\tilde{T}(\mathbf{l}) \tilde{T}(\mathbf{L} - \mathbf{l})}{K(\mathbf{l}, \mathbf{L})}$$



shift to larger
angular scales



For references, see e.g. Lewis & Challinor

How to measure CMB lensing ?

CMB lensing reconstruction

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WITH CMB
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In general, for two CMB fields X, Y , the estimators are of the form:

$$\hat{\phi}_{XY}(\mathbf{L}) = N_{XY}(\mathbf{L}) \int \frac{d^2\mathbf{l}}{(2\pi)^2} \tilde{X}(\mathbf{l}) \tilde{Y}^*(\mathbf{l} - \mathbf{L}) g_{XY}(\mathbf{l}, \mathbf{L})$$

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For the case $X = E, Y = B$, the minimum variance estimator has $g_{EB}(\mathbf{l}, \mathbf{L}) = \sin(2\varphi) \frac{\mathbf{l} \cdot \mathbf{L} C_l^{EE} + (\mathbf{L} - \mathbf{l}) \cdot \mathbf{L} C_{|\mathbf{L}-\mathbf{l}|}^{BB}}{\tilde{C}_l^{E,\text{tot}} \tilde{C}_{|\mathbf{L}-\mathbf{l}|}^{B,\text{tot}}}$

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The integral is a harmonic-space convolution $\implies$ Write as real-space product

$$\hat{\phi}(\mathbf{L}) = -N(\mathbf{L}) \int d^2\mathbf{x} e^{-i\mathbf{L} \cdot \mathbf{x}} \nabla \cdot [F_1(\mathbf{x}) \nabla F_2(\mathbf{x})]$$

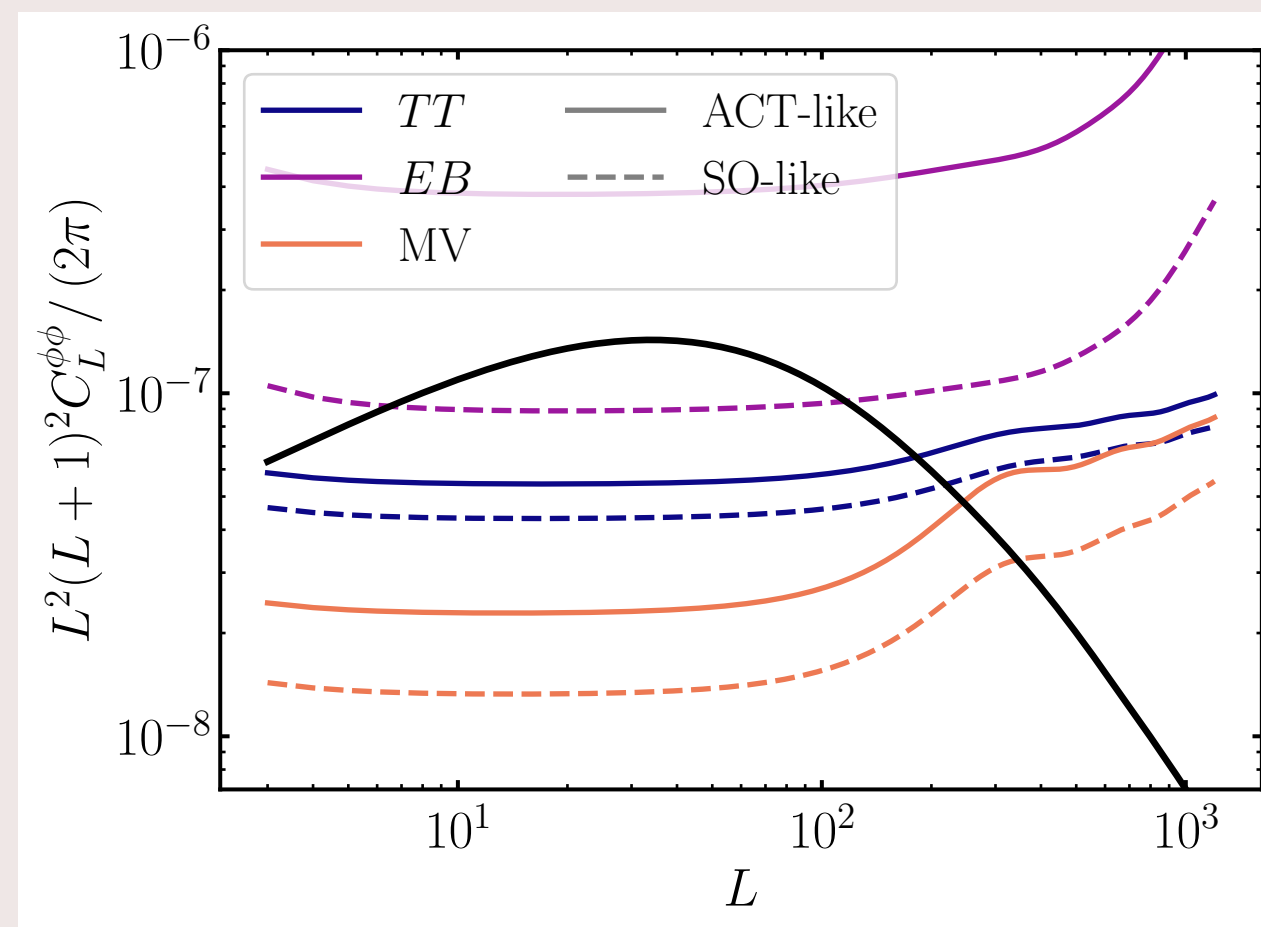
The CMB lensing power spectrum

Not as straightforward as one would imagine

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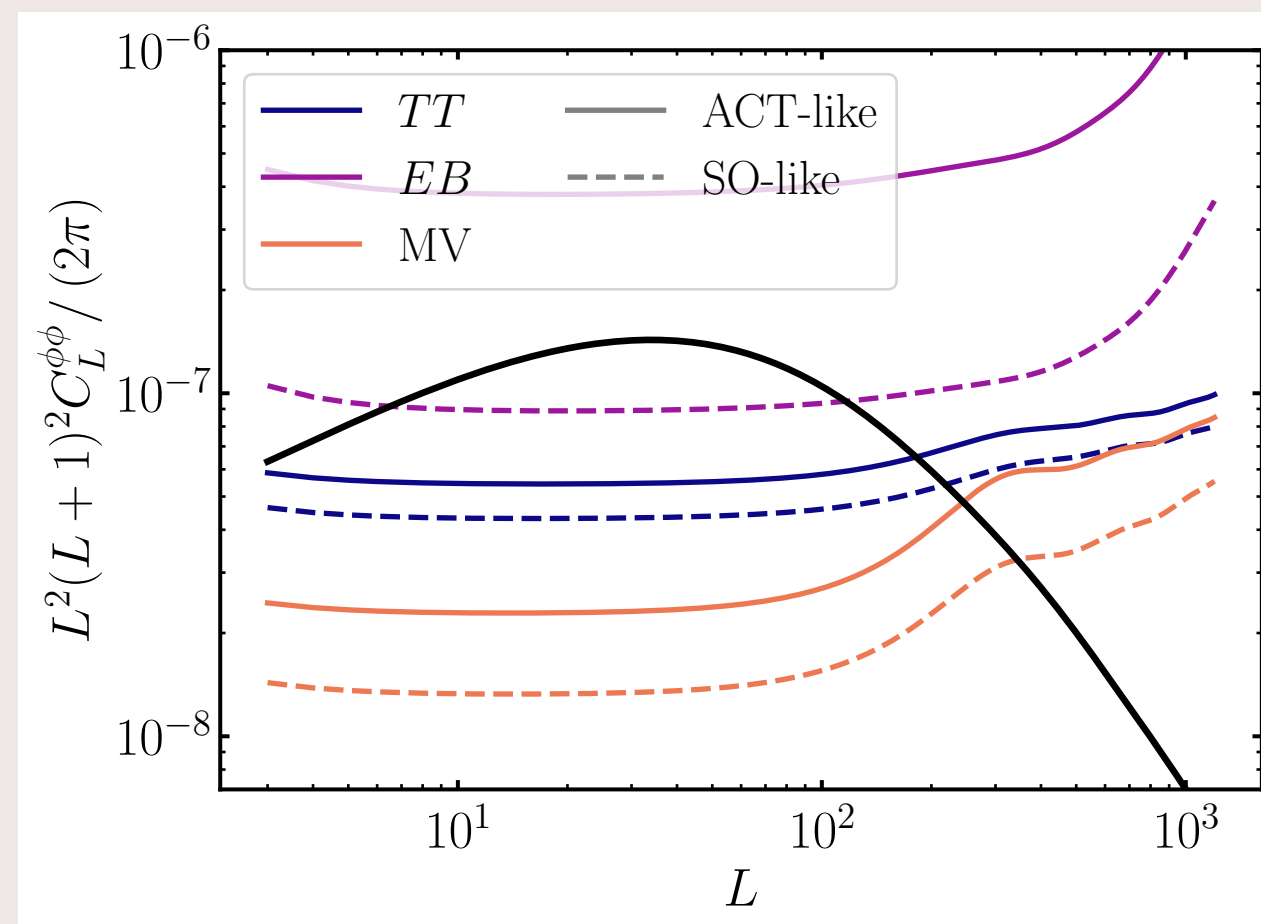
What is $N(\mathbf{L})$?



The CMB lensing power spectrum

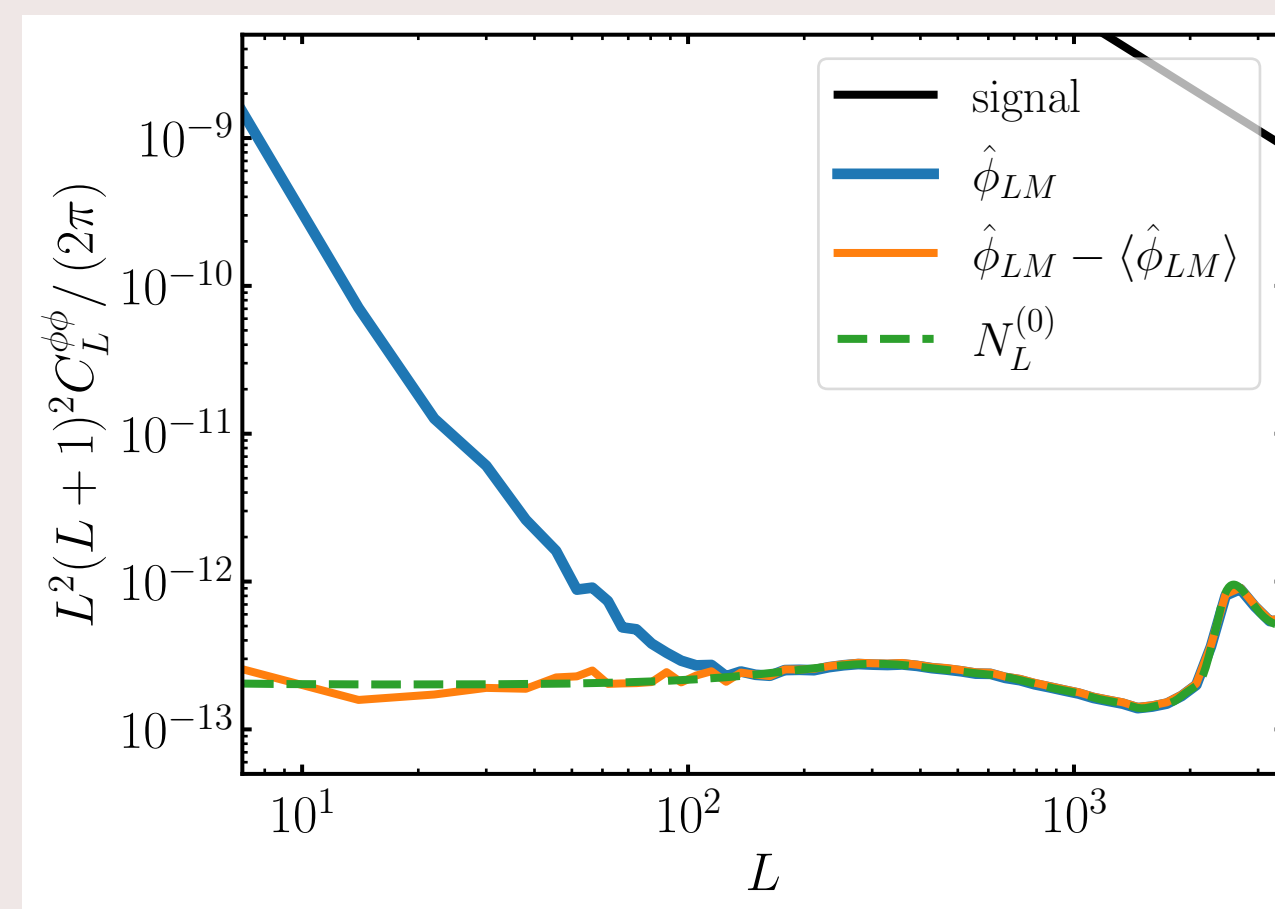
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How come I pick up a signal when I try to measure lensing on the unlensed CMB?

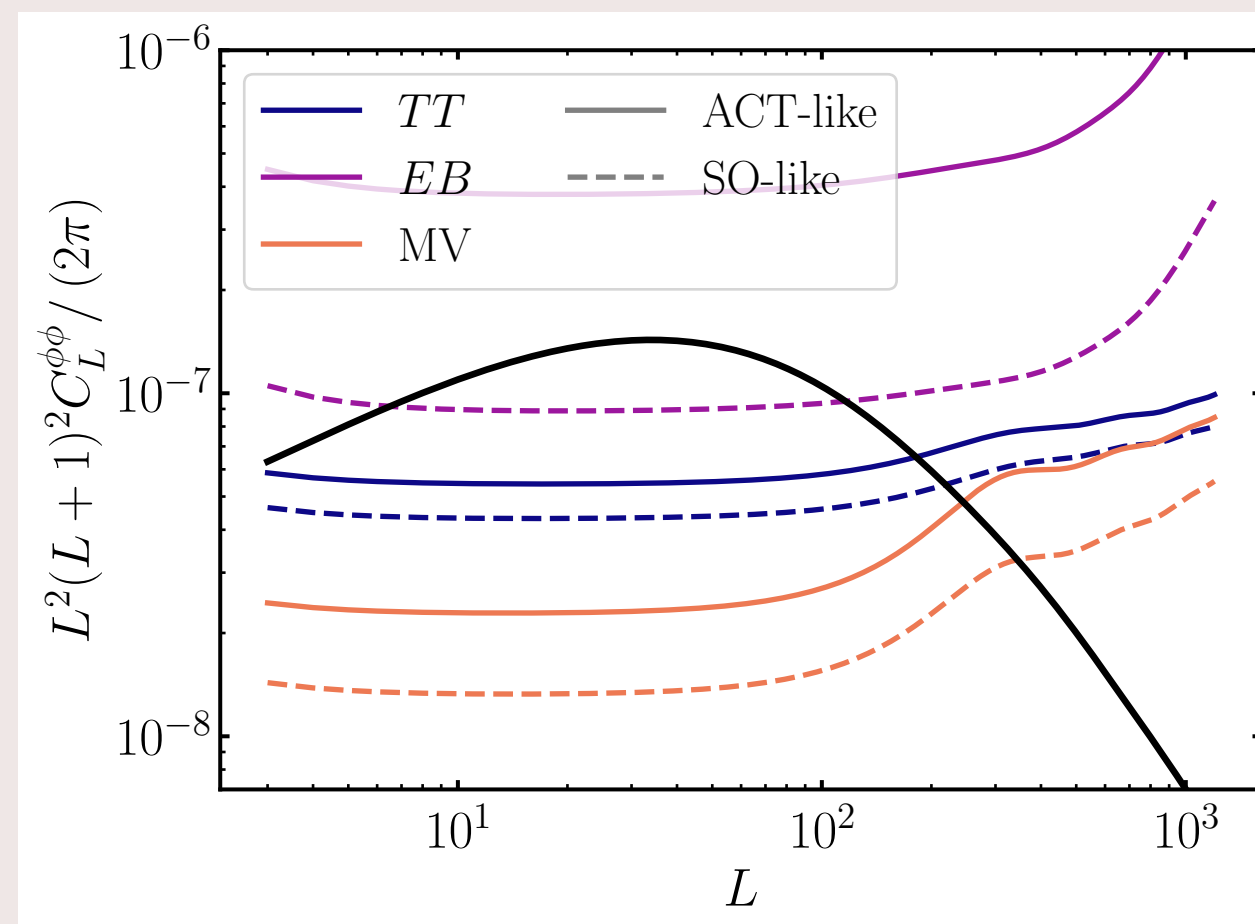
What happens in the presence of a mask?



The CMB lensing power spectrum

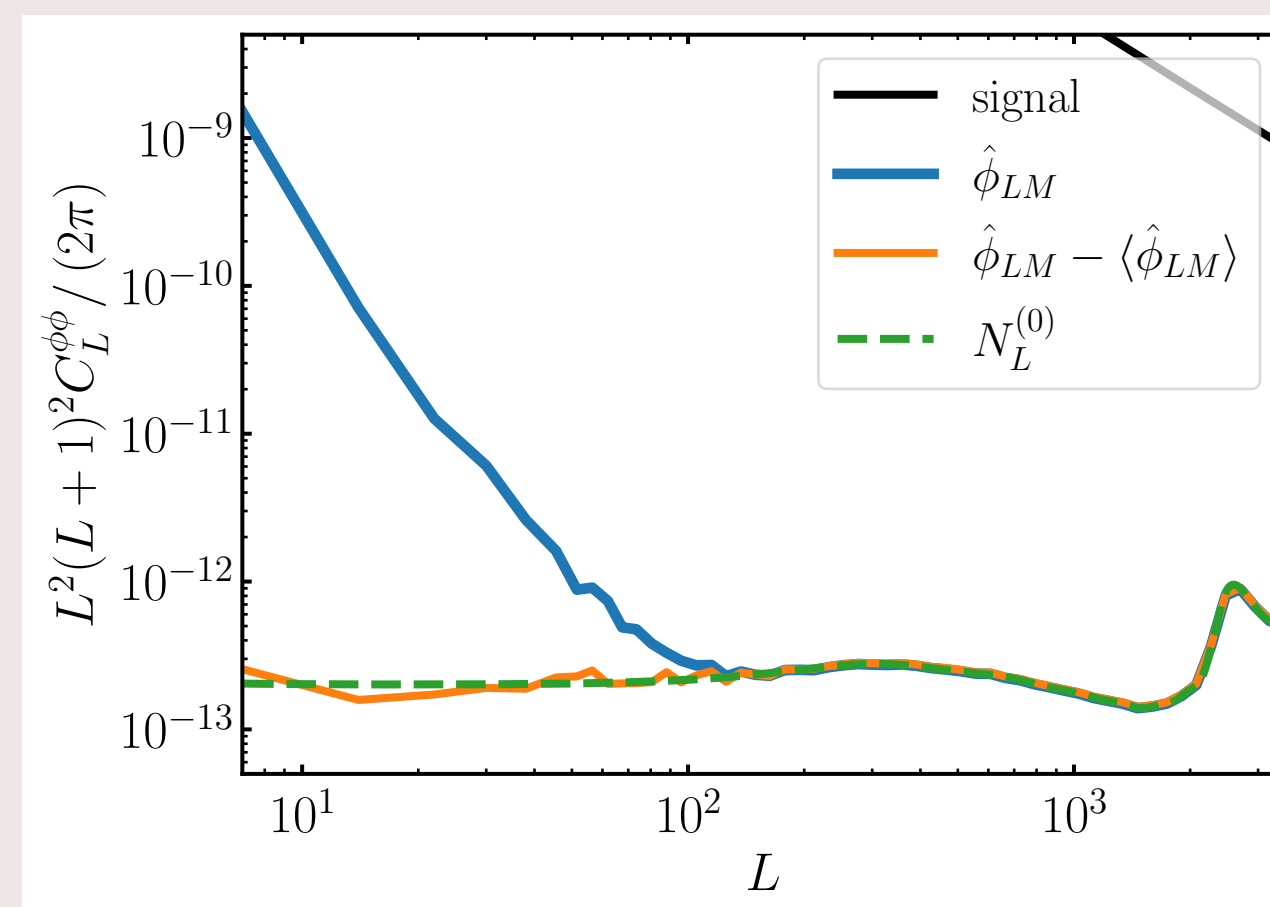
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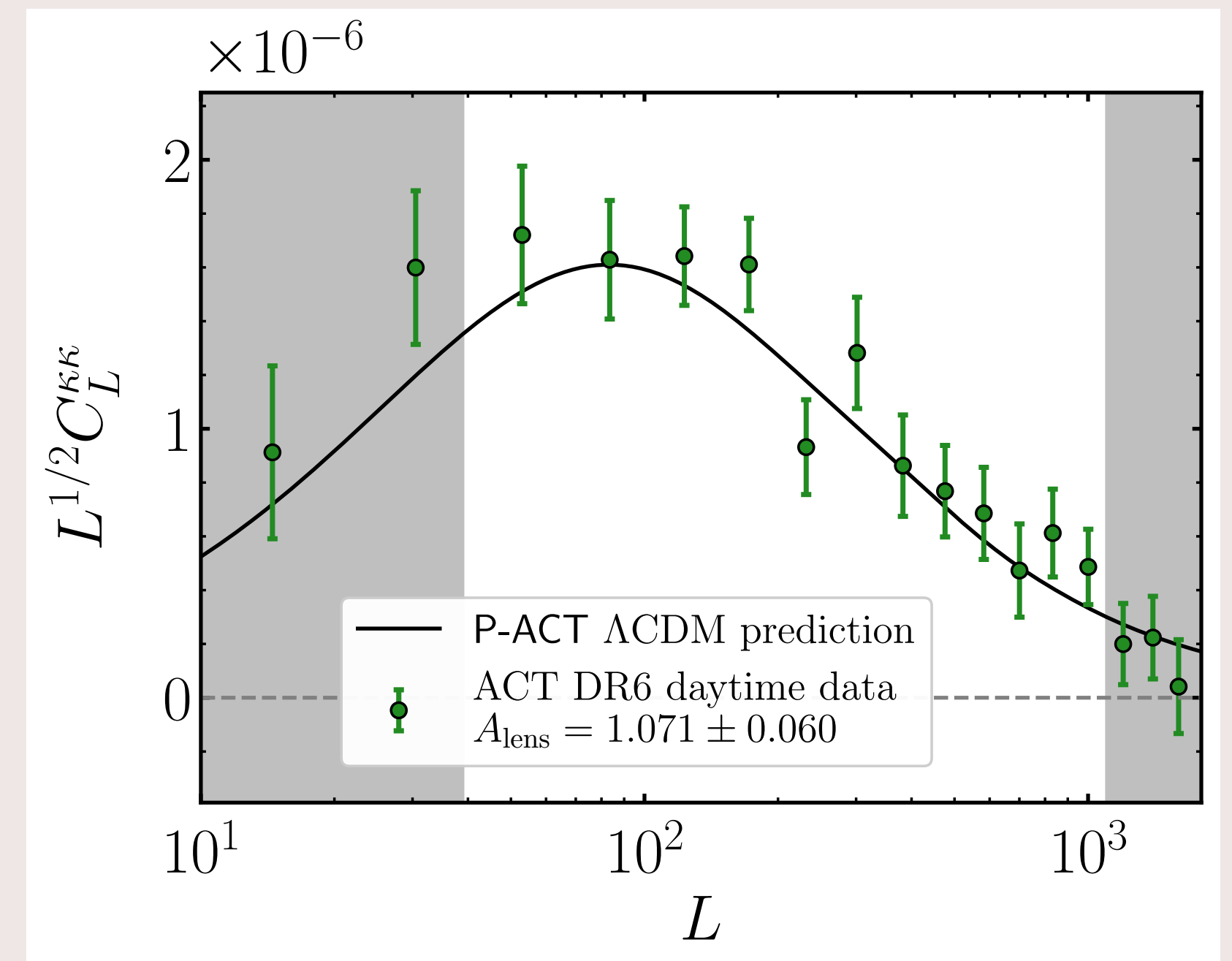


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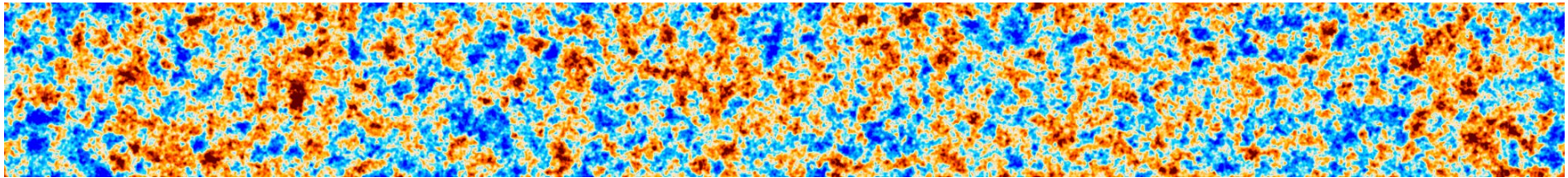
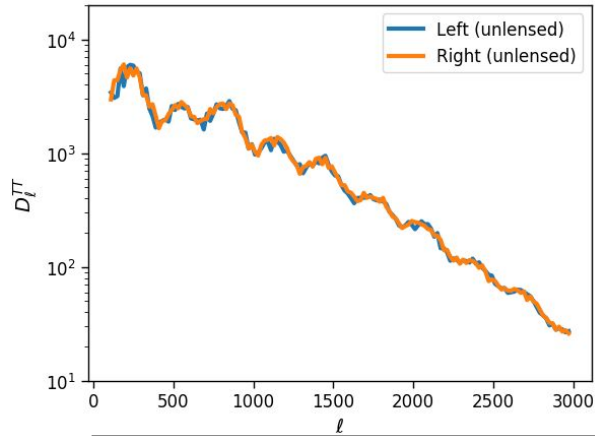


Lensing amplitude with respect to Λ CDM prediction (A_{lens})?



Statistical anisotropy from lensing

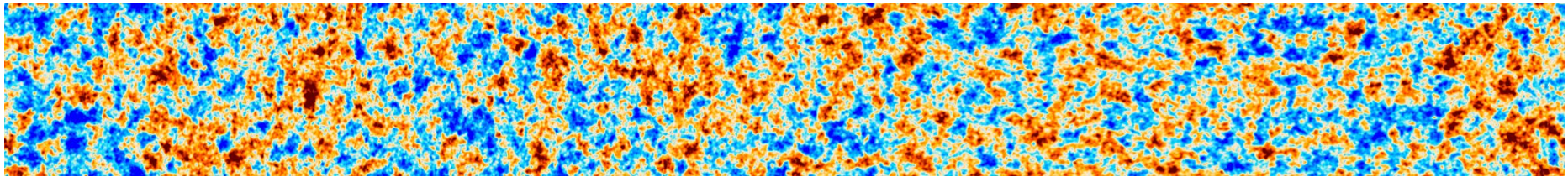
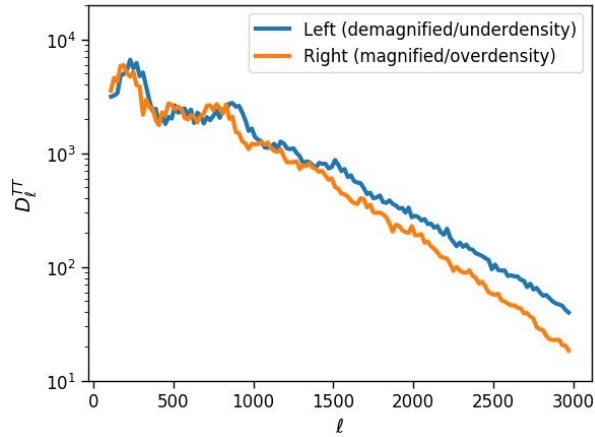
Matter overdensity mode
(zero)



Unlensed CMB
temperature

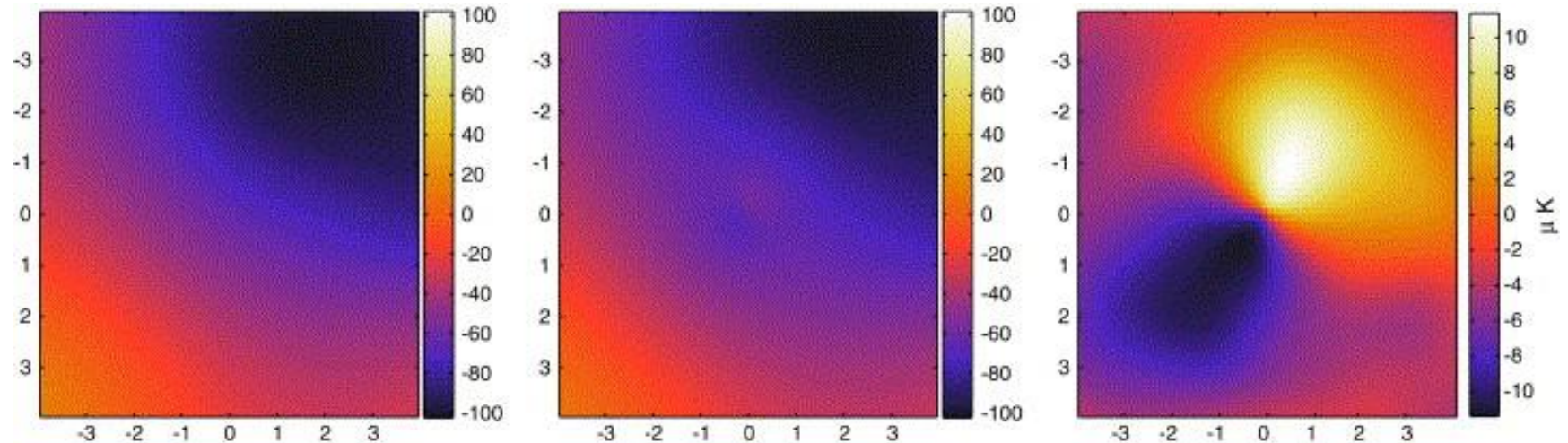
Statistical anisotropy from lensing

Matter overdensity mode



Lensed CMB
temperature

Antony Lewis, Anthony Challinor, Weak gravitational lensing of the CMB, Physics Reports, Volume 429, Issue 1, 2006, Pages 1-65, ISSN 0370-1573, <https://doi.org/10.1016/j.physrep.2006.03.002>.



Simulated effect of cluster lensing on the CMB temperature. Left: the unlensed CMB; middle: the lensed CMB; right: the difference due to the cluster lensing. In the middle figure note the direction of the gradient inverts inside the Einstein radius where the lensing deflections cross the centre of the cluster.